HORIZON-CL6-2022-ZEROPOLLUTION-01-04 - Securing drinking water quality by protecting water sources against pollution, providing innovative monitoring and treatment solutions and ensuring safe distribution

H_2OforAll

Innovative Integrated Tools and Technologies to Protect and Treat Drinking Water from Disinfection Byproducts (DBPs)

GA 101081953

Annex – Part B

Table: HYSTORY OF CHANGES

DATE/SECTION	NATURE OF CHANGE AND REASON
21.06.2022/Part B Section 1.1.1	The following passage has been added in Section 1.1.1 "behaviour over time and interaction with natural materials and micropollutants in freshwater as well". This passage outlines how the existing methodology and concept will address other pollution threats like micropollutants in response to a shortcoming from the ESR.
21.06.2022/Part B Section 1.2	The following passage has been added in Section 1.2 "The measurement accuracy of each sensing device …. . The above will increase and ensure the general safety of the sensing infrastructure, as well as the reliability and the reproducibility of the data." This passage outlines the methodology applied to deal with AI-based technical aspects in response to a shortcoming from the ESR.
21.06.2022/Part B Section 3.1	The Gantt Chart was updated with the deliverables due dates.
21.06.2022/Part A WP2	The following passages were added to WP2 description "…(D2.1)…"; "The selection of best location to place the sensing infrastructure along the water distribution network studied in Task 2.6 (Coimbra´s water distribution network managed by AC partner) will be based on the algorithms and methods developed in Task 2.4 (D2.2.)."; ". The algorithms and methods developed in this task (D2.2) will allow choosing the proper placement of the infrastructure developed in Task 2.2 in the pilot site of Task 2.6 (Águas de Coimbra)."; "…(D2.3)…" The data that will be obtained from the real water network monitoring will be important for WP5, WP6 and WP7. A final report will be made available with the lessons learned (D2.4).". These passages highlight the methodology applied to deal with AI-based technical aspects as well as the links between the tasks and the different WP in response to a shortcoming from the ESR.
21.06.2022/Part A WP3	The following passages were added to WP3 description "…and other pollutants (considered in Task 3.2 – see 3.2.1)…"; "D3.1 will consist in a report regarding the fate and behaviour of DBPs in the pipelines and the latter will be important to be given as input to Task 2.3". These passages outline how the existing methodology and concept will address other pollution threats like micropollutants as well as highlight the links between the tasks and the different WP in response to a shortcoming from the ESR..
21.06.2022/Part A WP4	The following passage was added to WP4 description "The formation of biofilm in the pipelines will be evaluated as a response to the different treatments to be applied. An experimental system for these tests will be built through a contractor with know-how for pipe-welding according to Atlantium design.". This passage highlights the need for subcontracting to build an experimental system for biofilm formation analysis. The following passages were added to WP4 description "The optimized adsorbents/solutions should be accomplished by M24, but the activities must continue until M36 in order to provide the necessary samples/methodologies for Task 4.2.4, Task 4.3 and Task 5.3."; The optimized MOFs/solutions should be accomplished by M24, but the activities must continue until M36 in order to provide the necessary samples/methodologies for Task 4.2.4, Task 4.3 and Task 5.3.; "…(Task 4.2.2 and Task 4.2.3),…"; "…(Task 4.2)…". These passages highlight the links between tasks in response to a shortcoming from the ESR..
21.06.2022/Part A WP7	The following passage has been added in WP7 description "…relevant on-going EU-funded projects (from Green Deal and HE projects funded in the call CL6-2022-ZEROPOLLUTION-01-04) and national/international programmes to take advantage of synergies and avoid work/topic overlap. Periodic meetings with sister projects of the same call (in a 6-month basis), joint communication materials, a joint policy brief (in collaboration with Task 6.3) as well as clustering activities such as the organization of a Water Projects Europe event will be developed and implemented.". This passage evidences the predicted clustering activities with sister projects
21.06.2022/Part A Deliverables	The following deliverables were added: - "D1.3. Data Management Plan – Updated"; "D1.4. Data Management Plan – Final". These deliverables consider the Data Management Plan update throughout the project. - "D3.5. Results of the hydraulic and quality modelling of water distribution network – Modelling Analysis". This deliverable was added as a sensitive version of D3.4 and will encompass confidential data from the drinking water consumers as well as private information from the company. - "D6.4. Final report on policy brief and recommendations".

	- “D7.2. Dissemination and Communication Plan – Final”. This deliverable considers the Dissemination and Communication Plan update at the mid-term of the project. - “D7.4. Final Exploitation Plan”. This deliverable considers the final updated exploitation plan. D2.1 type was changed to DEM since a sensing prototype will be achieved. D1.1; D1.2; D7.1 and D7.2 dissemination level was changed to PU. The following Deliverables delivery date was shifted as follows to align better with the reporting period: D1.6 (M6); D3.2 (M12); D5.2 (M34); D6.1 (M34); D6.2 (M34); D6.3 (M18) and D7.3 (M4).
21.06.2022/Part B Section 3.1	Table 3.1 Pilot Scale and demonstration summary table was added to highlight the scale, start and end TRL as well as the input and output materials of the demonstrator pilot scale units to be achieved during the project.
21.06.2022/ Part B Tables 3.1g	Table 3.1g was updated to provide clear insights about the activities to be subcontracted.
21.06.2022/Part B Section 4	Section 4. Ethics Section has been added including the proposal Ethics Issues Table as well as the following passage “Following the advice stated in the Ethics Summary Report, the Consortium acknowledges that some ethical issues might be faced regarding Humans, Personal Data, Non-EU countries and Artificial Intelligence. The Consortium confirms that compliance with ethical principles and applicable international, EU and national law in the implementation of research activities not originally envisaged (or not described in detail) in the DoA will be ensured. The Consortium also confirms that any ethical concerns raised by those activities will be handled following rigorously the recommendations provided in the European Commission Ethics Self-Assessment Guidelines.” This new section updates the Ethics self-assessment bearing in mind the advice stated in the Ethics Summary Report.
30.06.2022/Part A Critical risks for implementation	The following Risk was added in the risk log “Loss of an Associated partner (UK) organization (Low, High)” together with a mitigation plan. This new risk arises from the change of the UK partners statute from Beneficiaries to Associated partners.
27.07.2022/ Annex 2 Budget	There was a change in the budget allocation with 100.000 € being transferred from the partner Westus to the partner University of Twente with the following justification: In the initial proposal, University of Twente and Wetsus would like to use their joint H2OforAll budget to employ a full-time PhD researcher to carry out research tasks on behalf of both beneficiaries. This researcher would receive supervision, access to the research facilities, and other support from both University of Twente and Wetsus. However, formally it will be easier if the PhD researcher is hired only by the University of Twente. Thus, the researcher will be employed at University of Twente for the duration of the H2OforAll project but still perform some work at Wetsus (Westus will be a host rather than an employer). This does not change the global budget of the proposal.
29.08.2022/Part A Deliverables	D7.2 due date was changed to M20. D1.3 regarding the Data Management Plan update was removed. This changes were performed to align better the deliverables with the reporting period.
29.07.2022/Part B Section 3.1	The Gantt Chart was updated considering the changes in the deliverables .
29.08.2022/Part A Critical risks for implementation	The following Risk was added in the risk log “Wave of COVID-19 infections impose restrictions in presential work (Medium, Medium) together with the mitigation plan “Use of teleworking and reducing the traveling to the minimal required to achieve the project objectives. Use of rotative presencial work to keep the key experiments runing and obtain the results in due time, considering a plan for these experiments. Convert the planned project meetings to on-line meetings during restriction periods. ”. This new risk arises from the possibility of a new wave of COVID-19 infections during the winter.

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1. Excellence

1.1 Objectives and ambition

1.1.1 Overview and Objectives

The industrialization, agricultural activities and climate change are reducing the quality of drinking water sources. Thus, although the European Union policy to secure safe drinking water for all the citizens (Drinking Water, Bathing Water and Floods Directives) and the ongoing development and implementation of minimum requirements for water reuse, there are still persistent risks for human and ecosystems health. In fact, drinking water sources are often contaminated due to pollutants that are not removed by wastewater treatment plants or agriculture run-off. Moreover, the extreme weather events (that nowadays occur with increasing frequency, due to the climate change) are favouring the conditions for eutrophication (leading to algal blooms) and pathogens development in water sources. Lower quality water at the source implies harsher disinfection conditions, such as more intensive chlorination, to achieve the regulatory requirements of free chlorine and microbial parameters for drinking water. Disinfection byproducts (DBPs) are produced when chlorine reacts with organic matter in water during the traditional disinfection processes. The impact of DBPs at low concentrations on the environment and human health is still unknown. However, effects on liver activity and neurotoxicity were already detected when humans are subjected to large quantities of these compounds. Within this ambit, **H2OforAll aims to understand the impact of DBPs through a robust knowledge on their sources, pathways, occurrence, persistence and degradability, monitoring their incidence through innovative and robust systems, implement advanced preventive and mitigating strategies and measures avoiding their formation, remediate water containing DBPs through advanced and cost effective drinking water treatment processes and technologies and globally protecting the entire water chain while increasing EU guidance on measures to manage drinking water quality**.

More specifically, the **objectives of the present project** are to **(i)** assess main DBP sources and fate by developing suitable analytical techniques for monitoring such compounds and modelling their spreading through drinking systems; **(ii)** create a central knowledge base with the occurrence of DBPs in Europe; **(iii)** evaluate their environmental and health impacts while proposing measures to protect drinking water whole chain and soil (bearing in mind the role of soil health in the quality of freshwater); **(iv)** propose cutting-edge water disinfection technologies to prevent DBPs formation as well as treatments to remove them when their formation is unavoidable; **(v)** analyse life cycle and risks for the new technologies and materials proposed (adsorbents, catalysts, *etc.*); and **(vi)** increase public and policy-makers awareness for DBP issues by providing reliable data for new policy responses and guidance.

To successfully fulfil the overall objective, **several specific and quantifiable objectives** were defined (as described below), which have the indicated connection to H2OforAll Work Packages (WPs) architecture.

***O-I. Development of sensors and sensing infrastructures that allow identification of DBPs and/or the main materials they interact with high confidence. To enhance the sensing infrastructure with real-time monitoring capabilities and intelligently place it in the water distribution network* (WP2)**

Development of advanced, integrated, and **cost-effective water quality sensors for real-time, fast monitoring** and accurate analytical assessment of DBPs in water at very low concentration, **down to units of µg/L**. These methods will make use of sensors working in the IR domain, an enrichment membrane based on supramolecular interactions and a highly sensitive detector. These sensors will make part of a robust DBP monitoring system from the source to supply enhancing the sensing infrastructure with real-time monitoring capabilities bearing in mind also the main materials DBPs interact with. **Two kinds of field deployable IR-ATR analyser systems** are expected to be developed for water quality monitoring considering selected DBPs. Algorithms and methods will be developed to choose the locations to monitor the water distribution network wisely. **KPIs**: **At least six sensing devices will be strategically placed at AC partner** (end user drinking water distribution company) **network** (M2.1; M2.2).

***O-II. Development of real-time interactive maps of DBPs and development of a Central Knowledge Base* (WP2)**

Algorithms and methods based on high-likelihood spatial maps will be developed to select the locations required to monitor the water distribution network for DBPs. The data that will be gathered along the drinking water chain will allow to generate **real-time monitoring maps of DBPs from sources to supply**. A **Central Knowledge Base (CKB)** will be designed and developed to gather information about **DBPs distribution in time and space**. This platform will be **available for policy-makers, regulators and operators** of relevant infrastructures. The consortium involves 12 countries located all around Europe (including an end user partner) which will ensure that **DBPs concentration data from several countries will be available for feeding the Central Knowledge Base**. **KPIs**: CKB website up and running without failures during at least a year (M2.3; M2.4).

***O-III. Better understanding and assessment of DBPs formation and evolution* (WP3)**

Identification of the most important and hazardous DBPs and correlate their presence at different loads with risks for human health. Identification of gaps in legislation regarding drinking water both at the EU and global level. To understand **DBPs formation, behaviour over time and interaction with natural materials and micropollutants**

in freshwater as well as the distribution piping considering the materials and age. **KPIs**: This integrated knowledge will help defining the **best location for placing the sensing infrastructure** (connected with WP2) (M3.1).

***O-IV. DBPs toxicity and risk assessment analysis* (WP3)**

Multi-criteria decision-making techniques will be used to **prioritize target DPBs (20 DBPs will be selected based on their hazardousness)** that can be formed during disinfection (either by conventional or alternative disinfection technologies). For the most problematic DBPs, the **ecological risks** will be calculated based on predicted **toxicity data** as well as multiple bioassays. The data gathered will be used to develop new monitoring (WP2) strategies targeting the key contaminants and tools for disinfection/remediation (WP4) since it will allow to follow their efficiency regarding the toxicity of the by-products formed by these novel technologies. **KPIs:** Number of indices of water safety established and number of DBPs identified as hazardous (M3.2).

***O-V. Establishing preventive measures to protect drinking water* (WP3; WP6)**

To understand the **regulatory preventive measures in charge in EU and other countries** and identify lacunas regarding **measures and practices that might be added to EU regulation to protect whole drinking water** chain from sources to supply. These will also include **procedures to protect water sources considering extreme weather events**. Especial attention will be given to the revised EU Water Directive. **KPIs**: **A list of recommendations for safety planning will be devised and at least 3 policy briefs will be delivered (M3.3; M6.2).**

***O-VI. Development of new disinfection strategies for prevention of DBPs formation* (WP4)**

Development of **new disinfection processes and technologies to minimize/avoid the use of chlorine** thus **preventing/minimizing the production of DBPs**. This objective will be achieved by using under-explored features of **UV light** (simple or combined with intermittent chlorination). The new disinfection approach must **guarantee that the drinking water regulations are accomplished**. The by-products generated by the novel disinfection approaches will be followed using the analytical methods developed in WP2. Moreover, linked to WP3, the toxic effect of these compounds will also be obtained. **KPIs: New disinfection technology efficiency at least equal to the conventional method (M4.5).**

***O-VII. Development of advanced remediation technologies for DBPs removal from water* (WP4)**

Innovative and reusable adsorbents and catalysts will be developed to **sustainably enhance DBPs removal** through physicochemical processes. The **most efficient technologies**, bearing in mind DBPs removal capacity, toxicity abatement (linked with WP3) and reduced energy consumption **will be selected for further optimization**. **KPIs: Advanced remediation technologies efficiency and operating costs (M4.1; M4.2; M4.3; M4.4)**

***O-VIII. Integration of technologies for achieving safe drinking water* (WP4)**

The selected technologies will be integrated and further optimized using a pilot-scale process. The technology will be validated in a **lab-pilot scale facility operating with a water flowrate of 10 L/h**. **KPIs:** It is aimed to **reach the WHO recommended limits for DBPs** with the optimized process (M4.6).

***O-IX. Assessment of the impact of new technologies proposed for drinking water treatment compared to baseline technologies* (WP5)**

Assessment of **environmental and economic performance of the technologies** proposed and optimized in WP4. The costs and environmental performance will be **compared with baseline technologies**. **KPIs: Reduction on the environmental impact of at least 30% is expected.** The results will allow establishing recommendations for better sustainability performance of the new technologies. (M5.1; M5.2; M5.3).

***O-X. Raise citizens' awareness and engagement about environmental and health impact of DBPs, prevention measures within the household as well as from source to supply. Recommendations on prevention measures for safety planning. EU policy recommendations* (WP6)**

Actions for public engagement and awareness will be developed to inform about the sources of water pollution within and outside a household as well as **recommendations to prevent water pollution** and improve water quality. A **ranking list of prevention measures for safety plan** will be prepared and **delivered and reviewed by selected key stakeholders**. The relevant project results will be translated to **recommendations for policy adaptations at the EU level** and disseminated via wider outreach channels and direct communication with policy actors. Citizens' engagement actions will be performed by **using web platforms an open and interactive Water Museum** (partner AC). **KPIs:** At least 3 policy briefs delivered; number of participants at the engagement actions (M6.1; M6.3).

***O-XI. Dissemination and communication and exploitation of the project results* (WP7)**

Development of a **Dissemination and Communication Plan** (DCP) that will include channels and mechanism for successful implementation of dissemination and communication activities amongst the partners to the target stakeholders. Development of dissemination and communication materials and events while **promoting the engagement of relevant stakeholders and public** to maximize H2OforAll's impact. Ensuring the **project results exploitation** from a scientific, economic, policy and societal perspective. **KPIs:** Number dissemination and communication actions and participants (more details in Impact section) (M7.1; M 7.2; M7.3).

Table 1.1 shows the **alignment and pertinence of H2OforAll objectives regarding the work programme scope and highlighted points** in the text of the call HORIZON-CL6-2022-ZEROPOLLUTION-01-04. H2OforAll contribution to fulfil of the outcomes specified in the call topic are presented in Section 2.1.
It should also be noted that the focus of H2OforAll project and corresponding objectives avoid overlap with the targeted topics of previous Horizon 2020 and Green Deal calls, such as micro/nano-plastics, per- and polyfluoroalkyl substances, pharmaceuticals and contaminants of emerging concerns, pathogens and antimicrobial resistance. In addition, water reclamation and reuse or alternative water sources are not addressed in H2OforAll, as indicated in the HORIZON-CL6-2022-ZEROPOLLUTION-01-04 description.

Table 1.1 Contribution of H2OforAll objectives to the scope of HORIZON-CL6-2022-ZEROPOLLUTION-01-04.

HORIZON-CL6-2022-ZEROPOLLUTION-01-04 scope	H2OforAll Objectives
...expand the knowledge base required to identify, assess and prevent pollution threats (...) and the combined effects of multiple stressors on water sources, including risk assessment and management, to protect drinking water preparation and distribution. Particular attention to extreme weather events and possible synergistic effects affecting hydraulic flows, temperatures and pollutants' loads should be considered...	O-II O-III O-IV O-V
...development of sensors sensitivity, automated routine monitoring and fast analytical responses that fully integrate IT advances. ...extend the current analytical capacity to enable among other issues the detection of suspect and non-targeted pollutants, resulting in robust and reliable monitoring systems for consideration in future legislation. ...consider the requirements of the revised Drinking Water Directive...	O-I O-V O-X
Proposals in this field should further extend the knowledge base of mechanisms and reactions leading to DBP formation by analysing raw water quality and precursors, as well assessing drinking water treatment (DWT) operational parameters, including disinfection needs, methods and doses. Advanced DWT solutions should explore integrated systems coupling different treatment technologies and strategies enabling the optimization of the operational DWT effectiveness while removing DBP risks.	O-III O-VI O-VII O-IX
...the participation of academia, research organisations, utilities, industry and regulators is strongly advised, as well as civil society engagement whenever necessary, also aiming to broaden the dissemination and exploitation routes and to better assess the innovation potential of developed solutions and strategies.	H2OforAll consortium O-XI O-X O-IX

1.1.2. Going beyond the state-of-the-art and ambition

Disinfection by-products (DBPs) in drinking water have attracted wide attention and have become an important category of drinking water pollutants for regulation worldwide. During the disinfection process, disinfectants react with organic matter to produce a series of DBPs, including trihalomethanes (THMs), halo-acetic acids (HAAs) and haloacetonitriles (HANs) which can be carcinogenic, teratogenic, and mutagenic. H2OforAll is focused on understanding the emerging concerns regarding drinking water quality in relation to the disinfection operations and minimizing their impacts from the source to distribution. In this ambit, there is an urgent need to develop suitable monitoring systems across drinking network with the analytical capacity to detect suspected non-targeted pollutants (DBPs). An early-warning capacity will allow obtaining a robust online risk assessment system able to automatically detect eventual threats to human health. Besides, the DBPs behaviour along the drinking water distribution network and their interaction with naturally occurring materials and the pipelines (also considering their aging) must be well known to establish the placement of the sensing infrastructure wisely. We also aim to prioritize the target DBPs based on multi-criteria decision-making techniques by combining different methods and expertise based on toxicity data. These results will allow predicting the toxicity of DBPs produced by alternative disinfection technologies and approaches based on silico QSAR predictions. Combining the obtained results, preventive measures to protect the drinking water whole chain may be established and novel, integrated assembled solutions for alternative disinfection approaches as well as suitable technologies for DBPs removal from drinking water may be consolidated. The development of such alternative strategies implies the knowledge of the role of the raw water constituents on the DPBs production.
Table 1.2 shows the existing main limitations regarding DBPs monitoring, control, prevention, removal from the environment as well as the society and political awareness identified by the consortium, along with the ambition of H2OforAll to overcome them. A more detailed description of innovation and ambition of H2OforAll products/processes and methodologies is also given after this table based on the current state of the art.

Table 1.2 Main limitations identified for solving DPBs issues and the ambition of H2OforAll in this regard

Topic	Main Limitation	H2OforAll Ambition
DBPs monitoring	• Lack of cost-effective analytical devices able to determine DBPs along the drinking water supply chain • Intelligent placement of the sensing structure along the water distribution network	• Development of fast, cost-effective, and high-resolution analytical methods for DBPs based on hyperspectral imaging, and able to be applied in several media • Development of sensing infrastructures for DBPs inspection • Development of algorithms and methods for wisely determine the locations to monitor the water distribution network for DPBs
DBPs mapping and dispersion modelling	• DBPs dispersion and behaviour along the drinking water network is not known • Lack of European data base regarding DBPs	• Understanding how DBPs are formed, how they change over time and how they interact with natural occurring materials in freshwater • Build a Central Knowledge Base where information about DBPs can be added by authorized users to provide a global picture of DBPs in several European countries drinking water systems, considering the contamination change through time
DBPs toxic effects and risk assessment	• DBPs environmental impact, toxicity and risk not totally known (especially those produced by alternative disinfection technologies)	• Use multi-criteria decision-making (MCDM) techniques to prioritize the DBPs • Combine in silico QSAR prediction with MCDM to assist DBPs monitoring systems and optimize novel disinfection technologies as well as DBPs removal treatments
Innovative technologies for drinking water disinfection	• Water disinfection is mainly based on chlorination which leads to chlorinated DBPs • DBPs formation prevention and removal from disinfected water	• Development and analysis of new disinfection strategies by combining UV and intermittent chlorination and replacing chlorination by residual UV treatment • Development of advanced technologies aiming to remove the DBPs precursors from raw water to prevent DBPs formation. • Integrated strategies to remove DBPs from disinfected drinking water when their formation is unavoidable • Application of LCA/LCC assessment for comparing the environmental and economic performance of the proposed technologies compared with baseline technologies
Citizens and policy-makers engagement	• Lack of general public knowledge on sources of drinking water pollution and prevention measures • Lack of optimal prevention methods for safety planning • Lack of new policy guidance for EU policy makers	• Development of a digital platform and interactive actions to engage and create awareness to citizens on sources of water pollution within and outside the households • To select optimal prevention methods for safety planning • Create recommendations on prevention measures for safety planning of drinking water quality • Informing EU policy makers on project findings, preventive measures for safety planning and all technical project outputs

■ *DBPs Sensing, Analysis and Visualization*

The high concern around DBPs in drinking water requires the development of suitable monitoring systems giving reliable data regarding DBPs behaviour and fate along the drinking water wall chain. Only this way will be possible to establish suitable preventive and mitigation measures. Besides, such data is essential for imposing new regulation and policy making. In this context, there is a **need for analytical methods able to measure DBPs online even at very low concentrations in drinking water systems (µg/L).** A distinct goal of this project is the development of advanced optical sensing and monitoring systems that enable on-site and real-time detection and quantification of DBPs in surface, ground, and drinking water resources. State-of-the-art conventional analytical strategies require sampling followed by laboratory analysis, which is limited in response time, associated with high cost, and extensive efforts in sampling, sample transportation, and sample preparation/storage. Contemporary sensing systems mitigate some of these limitations, however, usually do not provide the required molecular discrimination. During the past decades, **optical sensing concepts have emerged as the most promising platform technology beyond the state-of-the-art for establishing field-deployable, compact, and molecularly selective sensing systems that may be continuously operated at harsh conditions**. Specifically, vibrational spectroscopic techniques with a particular emphasis on **infrared (IR) spectroscopy** have not only demonstrated their potential for identifying molecular

pollutants in real-world aqueous matrices at trace concentration levels[1], but also their capability of continuously quantifying the presence and dynamic temporal concentration fluctuations of persistent pollutants in a wide variety of aquatic systems[2]. In addition, IR spectroscopic sensing systems have evolved into compact and miniaturized tools that have already proven their capabilities during in-field operation[3]. Until recently, IR sensing technologies have been limited in sensitivity. This has been mitigated with the introduction of (a) signal amplification schemes based on so-called surface enhancement strategies, a.k.a., surface-enhanced infrared absorption (SEIRA) concepts, and (b) molecularly imprinted polymer and/or sol-gel enrichment membranes tailored for target DBPs. These concepts yield exceptional sensitivities after immobilization at or structuring of an appropriate waveguide surface, as used in the herein proposed infrared attenuated total reflection (IR-ATR) sensing strategies[4]. Consequently, **H2OforAll proposes to develop and deploy integrated IR sensing systems featuring (a) a microfluidic sampling interface taking advantage of SEIRA for ultra-trace analysis of DBPs, and (b) spectroscopic fiber optic probe for direct insertion**, *e.g.*, into a monitoring well or bore hole. Given the waveguide-assisted IR sensing technology, such sensing systems will be sufficiently **compact, portable and robust enabling usage in the field** (*e.g.*, at water monitoring stations) at locations of need provided by the partners in this project. Prototype system development will be augmented by the introduction of appropriate multivariate calibration and data evaluation/mining strategies using advanced AI and machine learning algorithms for correlating the obtained IR spectral datasets[5].

Ambition#1. The possibility of trustworthy and reliable low-cost, non-invasive techniques for the early detection of the quality of drinking water along the distribution network (from the source to the consumers) will revolutionize the management of water in cities, environmental policies and overall safety. In this project, a mid-infrared spectroscopic sensing system in a portable format will be developed for DBPs detection and quantification. H2OforAll consortium expects this technology to be mature in the next 5 years, *i.e.* by 2026, and aims for our solution to be one of the first complete ones available at the market by then. The possibility of detection and analysis via hyperspectral imaging at a low cost would be possible by then, which would make our case stronger and our models (based on new algorithms and methods considering the latest-technology machine learning and deep learning) much more accurate.

DBPs in drinking water come in many classes and are chemically diverse, which makes them challenging to monitor. Byproducts occur in presence of natural organic matter, anthropogenic contaminants (*e.g.,* pharmaceuticals), bromide, or iodide. To date, **more than 700 DBPs have been identified**. Hence, to **monitor DBPs and contaminants, samples (with low spatial and temporal resolution) are analysed in laboratories with expensive high-resolution analytical methods**, *e.g.* liquid and gas chromatography (LC and GC)[6]. These methods require proper sample preparation and analysis is costly and time-consuming. Special detectors are used to achieve the required resolution, *e.g.* LC-HRMS for broad spectral scanning, and GC-XSD for halogenated compounds.

To **reduce cost and increase the temporal and spatial resolution**, alternative methods are being researched. Fluorescence spectroscopy is an example providing sensitivity and real-time monitoring opportunities. As the spectra combine information of several components present in drinking water, artificial intelligence can be used to increase sensitivity and selectivity[7]. Another approach is to **combine data (data fusion) from existing water quality sensors and measurements (*e.g.* temperature, pH, UVA_{254} and Cl_2 concentration) to estimate DBPs[8]. This is very cost-effective and provides high spatial resolution and real-time monitoring**. These methods with artificial intelligence are still at their infancy and **require further optimisation to be applied in drinking water production.**

Ambition#2. Applying data fusion of data from all conditions (temperature, pH), joined with the required data infrastructure, to reach a digital twin of the drinking water treatment system. Obtain a cost-effective solution to monitor DBPs by combining a multitude of cost-effective existing water quality sensors, a smaller amount of higher resolution optical sensors, coupled to a validation in the laboratory with high-resolution analytical techniques reducing the cost of the monitoring program. Additionally, this solution will allow monitoring water quality with a much higher temporal resolution, so spikes in DPBs can be detected timely.

An important issue in this project is the **appropriate placement of the sensor devices that monitor water quality**, such that they achieve their predefined goal, which is covering the water distribution network as much as possible, maximizing the possibilities of capturing pollutants whenever and wherever they occur. This will involve two variants: (a) **maximizing coverage subject to a given number of devices, possibly with different capabilities (*e.g.* sensitivity) and sensing possibilities**; (b) **minimizing the cost of the sensory infrastructure given a minimally required percentage of coverage**. In literature, searching for the optimal location of sensors to cover some static or dynamic infrastructure has been approached via genetic algorithms[9] a Gaussian quadrature formula using the

[1] G.Dobbs, *et al.*, Anal. Chem. 79 (2007) 9566 ;[2] R. Lu, *et al.*, Sci. Rep. 3 (2013) 2525:; [3]C. Dettenrieder, *et al.*, Chemosensors 8 (2020) 75:1 ; [4]S. Repp, *et al.*, ACS Omega 5 (2020) 25036; [5] J.A. Seichter, *et al.*, Anal. Chim. Acta 1095 (2020) 48; [6] A. Andersson, *et al.*, Envir.Sci. Pollution Res. 26, 8 (2019) 7305; [7] M. Peleato, Sci. Rep. 12, 1 (2022) 1; [8]Z. Xu, *et al.*, Chemosphere 286 (2022) 131586,[9] R. Miller, Appl. Math. Comput. 97, 1 (1998) 71-97

functional gain as a weight function[9], an optimality criterion based on a measure of independence of the sensor responses in relation to the candidate locations[10], neural networks[11], eigenvector sensitivity analysis[11] and others. Kammer[14] proposed an effective independence algorithm based on the contribution of each sensor location to the linear independence of the identified modes. In this proposal, the problem of **optimized sensor placement will be approached in a more elaborate manner**, considering all relevant parameters and criteria, modelled in a 3D digital twin system, using an adapted and localized optimization function. **Solutions to the problem will combine geospatial analysis, geospatial probabilistic risk modelling, machine learning and modern AI techniques**, such as deep learning. Probabilistic risk modelling is a well-defined and understood technique used in risk modelling of infrastructures[15] as well as in disaster management and prevention in recent years[16]. From the modern AI techniques, we will exploit the time and space correlations of data and measurements, employing LSTM and RNN models.

Ambition#3. Go beyond the state of the art (SoA) via a more complete modelling and optimization process, employing digital twins, modern artificial intelligence (AI) and geospatial probabilistic risk modelling. This will lead to algorithms and methods for wisely choosing the locations to monitor the water distribution network for DBPs to obtain data to better understand their behaviour along the water distribution system.

A **global central knowledge base is lacking combining information regarding DBPs in drinking water around Europe**. This information combining the geographical space and time behaviour of DBPs will target policy-makers, regulators and operators of relevant infrastructures allowing to visualize the occurrence of DBPs in the countries of Europe. Moreover, the **interaction of DBPs with naturally occurring materials in fresh water as well as with the material pipes (bearing also in mind their aging) will be assessed by collecting data of 12 months of monitoring**. This information will be crucial to intelligently define the monitoring infrastructure along the water distribution network (AC partner will provide the case study infrastructure), as well as the **needs for further disinfectant injection along the distribution to ensure the legally required residual disinfectant load**. In addition, a user-friendly and smooth experience for the operators can improve performance and encourage the duty. The development of a central knowledge and data base will be carried out, in correlation with the goals established in the proposal. A **digital tool will be developed in H2OforAll to cover the recollection of technical information regarding some specific contaminants in urban water streams** to provide real evidence of the contaminant's presence and the efficiency of the solution treatments implemented within the project. The digital tool planned for this project will be built in a multilayer format and will include a number of specific architectures and modules, *i.e.* data acquisition layer, data treatment and data storage layer and user interface. The information collected and introduced to the actual platform will be collected from the net of sensors and structures also designed and implemented in the project. The digital platform components will be customized and optimized as the function of the data gathered as well as the actual sensors implemented, allowing to monitor the water quality and the selected parameters.

Ambition#4. To build a central knowledge base with DBPs occurrence information in European countries (that can be spread worldwide). This will allow visualizing DBPs from sources to supply in a real-time interactive map. The information will be made available to operators and regulators indicating the location, date/time and pollutant time allowing a more suitable monitoring of drinking water quality. Moreover, aggregated/summarized information will be also made available to the general public without exposing the private information of local operators. This will raise the public awareness on the DBPs issues and will also be a driving force for policy-making and regulations.

■ ***Environmental Impacts and Risk Assessment of DBPs and Prevention Measures Analysis***

In response to the health risks, some countries and authorities, such as the United States, China, and the European Union, have regulated DBPs thresholds in drinking water. **Over 700 DBPs have been identified so far, but only few of them are regulated internationally.** Moreover, **previous research works focused mainly on the health risks of DBPs in drinking water**. In comparison, **little attention has been paid to the risks of DBPs in the environment**, with no consideration for whether they are intentionally released from a point source (*e.g.*, lakes, water ponds, springs) or unintentionally discharged from a non-point source (*e.g.,* farming). Also, the **intensified disinfection of domestic water during the COVID-19 pandemic increased the release of toxic DBPs.** Though the toxicity of most of the DBPs is evaluated using instrumental methods of analysis and Quantitative Structure Activity Relationship (QSAR) tools, it can be still difficult to identify the compounds that should be considered environmentally hazardous, given the large quantity of data available [16]. One solution is to **use tools that allow the ranking of compounds according to different defined endpoints**. One of these tools is **Toxicological Priority**

[10]A. Wouwer, *et al.*, J. Process Control 10, 4 (2000) 291; [11] K. Worden, *et al.*, Eng. Struct. 23, 8 (2001) 885; [12] Z. Shi, *et al.*, J. Eng. Mech. 126, 11 (2000) 1173; [13] D. Kammer, J. Guid. Control, Dyn. 14, 2 (1991) 251; [14]J. Sadler, *et al.*, IET Intell. Transp. Syst. 10, 1 (2016) 17; [15] S. Kumar, *et al.*, Cold Reg. Sci. Technol. 165 (2019) 102813.
[16] M. Eid, Journal of Polish Safety and Reliability Association 1 (2010) 83–87.

Index Graphical User Interface (ToxPi GUI) software. ToxPi is a flexible decision-support tool that allows the integration of different metrics established for each DBP, providing a ranking of compounds by transforming the data into a dimensionless score index, with different scales and attributing a different weight to each endpoint. Tools such as ToxPi are based on multi-criteria decision-making (MCDM) techniques that treat endpoints and their relative importance as criteria and weights, respectively, and the compounds as alternatives to be compared for ordering or selection. **TOPSIS (Technique for Order Preference by Similarity to Ideal Solution)** is an MCDM technique that ranks alternatives in relation to the positive-ideal or the negative-ideal, defined from previously established criteria and weights, considering the characteristics assessed. However, until now, **no studies have used ToxPi and TOPSIS to rank DBPs with higher environmental risk. In addition, no reports were found in the literature that have compared screening results obtained using different MCDM methods.**

The risk analysis is aimed at assessing whether the system can provide water that meets health standards in accordance with applicable water quality standards for human consumption. The evaluation of risk is a strategy suggested as a parallel approach to the validation of water by water-quality norms. **Risk determination is a trend that can be implemented in water safety plans** and evaluate new strategies in the security production and management process of the water supply systems.

Ambition#5. Shade light about the hazardousness of the different DBPs by ranking the main compounds according to different defined endpoints. Multi-criteria decision-making (MCDM) techniques will be used for the first time to rank the compounds. The toxicity data and the toxicity category of the 20 DBPs that will be selected as a reference will be applied to predict (through silico QSAR predictions) the toxic character of new DBPs that may be produced when novel water disinfection/treatment strategies are applied as well as the compounds that may be produced by the initial DBPs decay along the drinking water distribution network. Thus, this will assist the development of both the monitoring and disinfection/remediation technologies.

Nowadays there are no regulation nor guidance for preventive measures to be applied in private houses to protect drinking water (from sources to supply). H2OforAll will make efforts to understand regulatory preventive measures needed and used in other countries to **establish as much as possible preventive measures and practices to ensure quality drinking water and mitigate problems along the whole chain**. The application of **drinking water periodic surveys and risk assessment and management is new in the EU**. Within this methodology several preventive measures can be defined. Moreover, the preventive measures and practices that will be collected will also aim to **reduce the contamination at the drinking water source so that intense disinfection is not required** leading to the formation of lower amounts of DBPs. This will allow establishing procedures to **protect water sources bearing in mind also extreme events related to global climate change**.

Ambition#6. To understand preventive measures and regulations for drinking water protection applied in other countries and establish novel methodologies to protect the whole chain of drinking water. By understanding the role of the raw water in disinfection demand, to stablish preventive measures to protect drinking water sources from pollution to reduce the need of for intensive disinfection.

■ ***Advanced and Cost-Effective Treatment Technologies for Drinking Water***

To **replace or minimize the usage of chlorination** in water distribution systems, some **under-explored features of polychromatic UV** (emitted by medium pressure [MP] lamps) will be studied. Previous works demonstrated the residual effect of MP UV in desalination settings, but not in water distribution systems, where water may travel for many hours from the source to the user. The **residual effect of UV may act directly by long-term damage to viable bacteria, or by generating stable radicals**. To determine the mechanism by which the residual effect is generated, different types of MP lamps will be tested in the presence or absence of compounds that may serve as precursors for radical generation.

Ambition#7. To develop innovative disinfection technologies that can avoid/reduce the chlorination minimizing this way the production of DBPs. Novel polychromatic UV disinfection strategies will be studied combined (or not) with chlorination whenever necessary. The potential of the UV residual effect will be explored by using different lamps in the presence or absence of radical precursors. The ambition will be to reduce the application of chlorination diminishing this way the amount of DBPs formed.

The temperature increase due to climate change enhances the production of DBPs since the reaction rates are higher. Moreover, it reduces the dissolved oxygen at the water sources[17] which promotes algal blooms that are precursors of DBPs[18]. Another source of DBPs is natural organic matter (NOM) present in the waters as well as contaminants from

[17]Brown and Caldeira, Nature 552 (2017) 454-50; [18] B. Walling, *et al.* Environ. Monit. Assess. 189 (2017) 225; [19] M. Zazouli, *et al.* J. Environ. Health Sci. Eng. 15 (2017) 25;[20] X. Sun, *et al.* J. Environ. Sci. 81 (2019) 52-67; [21] J. Sánchez-Polo, *et al.*, J. Colloid and Interf. Sci. 300 (2006) 437; [22] F. Wang, *et al.*, ACS Nano 14, 7 (2020) 8975-8984; [23] P. Punyapalakul, *et al.*, J. Hazard. Mater., 171 (2009) 491-499; [24]H. Furukawa, *et al.*, Science 329 (2010) 424-428 ; [25] X. Zhang, *et al.*, Chemosphere 242 (2020) 125144 ; [26] K. Adegoke, *et al.*, Mater. Chem Physics 252 (2020) 123246; [27] R. Li, *et al.*, Adv. Mater. 30 (2018) 1705512

diffuse and point sources of water pollution that damage the quality of drinking water sources. Thus, **technologies must be developed to reduce raw water contamination before the disinfection which will reduce the DBPs produced**. Moreover, **treatment processes shall be developed to remove DBPs when their production is not avoided** at the disinfection step. The adsorption and membrane filtration have been used as an alternative to reduce the DBPs but with low efficiency[19]. Alternatively, to minimize the DBPs production their precursors (such as NOM) may be removed before disinfection[20].

The **literature about the cleaning of drinking water through adsorption on aerogels is almost inexistent**. Only two works were found related with this subject, one of them[21] using Ag-doped activated carbon aerogels for bromide and iodide removal from drinking water, and the other[22] developing super-elastic skeletal-structured silica nanofibrous aerogels with the rechargeable bactericidal and antifouling properties through N-halamine moieties grafted in electrospun silica nanofibers. While the first work aimed at drinking water cleaning, the second one aimed at its disinfection. However, the method involving N-halamine still promotes the disinfection by chloride effect, claiming the authors a rechargeable chlorination capability of >4800 ppm. Thus, **great advances may be driven in this field through the design of aerogels that can remove DBPs from drinking water**. In fact, aerogels are highly porous (>90 vol%) and versatile materials which makes them excellent candidates for enhancement of surface-dependent processes such as catalysis and sorption. Silica-based aerogels can also easily have their surface chemistry tuned due to the large panoply of silicon alkoxides available in the suppliers. **Organically modified silica (ORMOSIL) aerogels will be prepared in H2OforAll with selected functional surface groups for capturing DBPs**, *e.g.* amine and thiol, and also counting on the OH groups inherent to silica aerogels. These moieties on mesoporous silicates already proved to be promising[23] in the adsorption of dichloroacetic acid, being the interactions explained by electrostatic forces and hydrogen bonding.

On the other hand, **metal organic frameworks (MOFs)**, as inorganic-organic hybrid crystalline materials, possessing high specific surface area (up to 6240 m^2/g as MOF-210 reported in *Science*[24]), structural diversity, controllable pore sizes, recyclability, numerous active sites, high tunability and tailorability and excellent selectivity as an adsorbent, are **potentially attractive materials for both pollutants' separation and photocatalytic oxidation**[25]. MOFs can be diversified by varying the metal ion centres and organic linkers and simple altering the MOFs network could result in their functionality modification and the lengths of its spacers. The **uniqueness of MOFs as adsorbents is due to their structural diversity and easy of structural modification** of both the inorganic components and the organic bridging ligands in changing them from being microporous to mesoporous materials. Adsorption capacities of previously reported MOFs reached value 703 benzene mg/g and 1188 nitrobenzene mg/g for MOF-5 and MIL-68(Al), respectively, and it was revealed that MOFs perform better than absorbents like Zeolite and activated carbon[26]. Moreover, modification of MOFs by functional groups could enhance their selectivity and adsorption capacity towards desired pollutants. **Semiconducting MOFs** undergo a similar charge-separation process to that occurring in TiO_2 (or other metal oxide photocatalysts), but the VB and CB are defined as the highest occupied molecular orbital (HOMO) and the lowest unoccupied molecular orbital (LUMO), respectively, according to the molecular orbital method. Commonly, the HOMO energy level of the organic linker is regarded as that of the MOF, while the redox potential energy level of the metal–oxo cluster is regarded as the LUMO energy level of the MOF[27]. Therefore, oxidation reactions take place in organic linkers, while reduction reactions take place in metal–oxo clusters. Thus, photocatalytic properties (*e.g.* oxidation potential, reduction potential and band gap) could be easily customized by playing with medal ions and organic ligands.

Considering circular economy strategy, the **application of biochar coming from different feedstocks** (wood, demolished wood, garden waste, sewage sludge, food rest biowaste, cardboard waste) will be **innovatively evaluated for DBPs and DBPs precursors removal from water**. To improve affinity towards the contaminants, **biochar modifications** will be targeted using clay, polymers, metal-organic frameworks (MOFS), as well as magnetic bio-activated carbons to facilitate their separation from the liquid phase.

AOPs for water treatment, like ozone-based processes, Fenton-like and photocatalysis have already demonstrated their contribution to minimize pollutants from water when properly integrated in drinking water facilities, as well as in municipal and industrial WWTP. **However, their in-situ application is not reported**. H2OforAll project seeks to deeper the study of several advanced oxidation processes, especially those involving **novel catalytic materials**, as promising and applicable technologies to be used in the **removal of DBPs and DBPs precursors from water protecting the whole drinking water chain** from the source to the supply. This will give reliable tools to face current and emergent water problems, enabling novel solutions to protect water bodies, aquatic ecosystems (and all the interacting ecosystems such as soil) and further enhancing water quality and its management for safe human and ecological use. Moreover, the **AOPs will also be innovatively applied in the regeneration of aerogel and biochar DBPs-loaded adsorbents**, allowing the mineralization of these pollutants and the reuse of the adsorbent.

Ambition#8. Implementation of advanced preventive and mitigating strategies and measures based on adsorption and advanced oxidation technologies applying novel materials to protect drinking water sources, treatment and supply against harmful effects bearing in mind global and climate change. The integration of technologies to reach a suitable sustainable, assembled strategy to efficiently treat raw water to remove contaminants that potentiate algal blooms and DBPs formation as well as DBPs removal strategies from disinfected water will be an asset of this project. A lab-pilot unit will be designed to validate the selected treatment approaches using real treatment water conditions.

■ *Environmental and Economic Sustainability Assessment*

For a **fair comparison of the environmental performance of DBPs prevention and removal technologies** with chlorination as their conventional counterpart, **DBPs residues should be accounted in the environmental assessment**. However, choosing water quantity as a functional unit is a common approach in LCAs of drinking water production technologies[28] . Even the LCA studies that compare technologies aiming at removal or reduction of DBPs[29] use a specific amount of produced water as a comparison basis. **Only a few studies like Amini *et al.*[30], includes water quality as well as water quantity in the assessment**. Their results show that, this difference in accounting water quality (in their case dissolved organic carbon (DOC) concentration), causes a dramatic difference on the results. However, they only conducted this assessment for ion exchange treatment, and DOC concentrations are only one of the precursors of DBPs formation. In this study, a functional unit accounting for the water quality in terms of DBPs will be proposed based on experimental studies conducted throughout the project. The **impact of the choice of different functional units (accounting and not accounting for the water quality) on LCA** results will be also investigated to propose best practices for environmental assessments in the following studies.

Ambition#9. To select wisely innovative processes for water treatment and disinfection using Life Cycle Analysis and Life Cost Analysis to compare them with the conventional techniques. This will innovatively involve not only the water quantity produced as functional unit but most importantly water quality.

■ *Guidance on Measures, Policy and Public Engagement*

The awareness and engagement of civil society is key to preserve the quality of water. On the one hand, **citizens experience severe health and environmental risks due to DBP pollutants and they often lack information** about what these pollutants are, what risks DBPs entail and even what they can do about it. Water quality is taken for granted and it is extremely **hard for the households to monitor or influence the water quality they experience**. For instance, **sensory and treatment technologies are often inaccessible or costly for end-consumers of water**. On the other hand, **citizens' themselves are often the source of water pollution**. Household activities such as laundry that involve detergents can chemically contaminate the disposed water, which later makes the use of treatment technologies imperative causing new contamination by DBPs.

Ambition#10. To engage and create awareness to civil society to the health and environmental risks associated to DBPs. Moreover, to raise public concern to the fact that water quality is not granted and that water consumers are also sources of water pollution. This will be integrated with preventive measures to protect the water that can be used in our everyday life targeting a zero-pollution environment.

■ *Dissemination, Communication and Exploitation*

H2OforAll's communication and dissemination activities will engage a variety of stakeholders – researchers, universities, companies, associations, governmental organizations, and other entities in the scientific and technologic ecosystem working in water quality. Given the geographic dispersion of the partners in and outside Europe, the project will **use its global networks and communication channels to ensure widespread dissemination**.

These activities will disseminate the project findings widely in an accessible manner, including through the organization of workshops and the promotion of co-creation activities amongst relevant stakeholders (consumers, policy makers, researchers and entrepreneurs), as well as the publication of useful materials such as free educational and research materials and publications related to DBPs and water quality.

H2Oforall's WP7 has been established to **disseminate enhanced knowledge on DBP spread, occurrence, persistence and degradability**; as well as **prevention and mitigation strategies**, thus contributing to a scientific implementation and dissemination of the **priorities established in the European Green Deal and SDG6 of UN**.

Ambition#11. Widely disseminate the results obtained in H2OforAll involving several publics (researchers, companies, associations, governmental organizations, and general public) to enhance their impact at several levels (scientific, societal and economic). To give a strong contribution on the dissemination of enhanced knowledge on DBPs spread, occurrence, persistence and degradability, preventive and mitigation strategies enhancing the competitiveness of the EU water sector while leveraging the EU's position and role in the global

[28] A.Bonton, *et al.*, Desalination 284 (2012) 42-54 ; [29] W.Mo, *et al.*, Water Res. 143 (2018) 155-164; [30] A.Amini, *et al.*, J.Clean.Prod. 104 (2015) 413-421

water scene. H2OforAll will contribute to a scientific implementation and dissemination of the priorities of the European Green Deal and SDG6.

1.1.3. H2OforAll positioning, from "idea to application"

Technologies and methods used in H2OforAll for monitoring DBPs in drinking water, water treatment & disinfection and stakeholders/citizens' awareness and engagement tools, show a good level of maturity – **Table 1.3**.

Table 1.3 H2OforAll TRL positioning.

Component / method / technology / process	Related milestone	Initial TRL	Final TRL
Online interactive CKB with information about DBPs in drinking water around Europe	2.4	N/A	6
DBPs sensing tools with low-cost spectrometers	2.1	3	5
Digital twin of the water distribution network for DBPs behaviour modelling/simulation	2.2	3	5
Alternative disinfection processes using MP UV lamp	4.1	3	5
Functionalized aerogels and ACarbon for organic pollutants and DBPs tailored sorption.	4.2	3	5
Catalytic ozonation & integrated ozone-treatments.	4.2	4	6
Application of heterogeneous photocatalysts for pollutants removal.	4.2	3	5
Citizens' awareness and engagement digital platform.	6.1	2	4

Most of them already **start from an experimental proof of concept (TRL=3), and only some very specific and innovative technologies start from a TRL=2 (technology concept formulated)**. The ozonation technology shows a higher starting TRL. The **final TRL of the technologies will be mostly between 5 and 6**, aiming at the **validation** (and in some cases demonstration) of the developed technologies in relevant water and drinking water treatment conditions. The sensors and monitoring infrastructure prototypes will also be validated in the field, with the collaboration of the end-user of the Consortium (AC).

1.2 Overall Methodology

H2Oforall methodology was exclusively designed to efficiently answer the specific challenge of the call: *Securing drinking water quality by protecting water sources against pollution, providing innovative monitoring and treatment solutions and ensuring safe distribution.* The applied methodologies to fulfil the specific objectives of the project are presented hereafter:

Methodologies to accomplish O-I: A mid-infrared (MIR) spectroscopic sensing system in a portable (*i.e.*, shoebox-sized) format for detecting DBPs in drinking water will be developed in H2OforAll. The system will comprise a mid-infrared light source with appropriate waveguide technology (*i.e.*, fiberoptic or hollow waveguide-based sensing interface) for detecting DBPs directly in aqueous matrices. This will be combined with selective enrichment materials augmented by molecular imprinting technologies to increase the sensitivity. The IR-ATR water analyzer will enable online and inline monitoring of dissolved pollutants near real-time without the need for elaborated, costly and time-consuming sampling and laboratory analysis. These new sensing technologies will be validated using the traditional analytical method (T2.1). These new sensors will be used to develop low-cost sensing devices and sensing infrastructures that will be used to identify and monitor the main DBPs that will be selected in WP3 and the main materials they interact with. The sensing infrastructure will be open in a do-it-yourself approach and each sensing device will be able to form a network with the rest of the devices to propagate measures through the network (T2.2). The measurement accuracy of each sensing device will be ensured by elaborate calibration processes while the measurement consistency of the sensing infrastructure will be assured by checking and comparing the data collected from neighboring devices. Data transmission will be based on modern energy-efficient encryption standards, ensuring data security, while near real-time topology control will make sure that no device is under attack or becomes faulty. When one device is manually

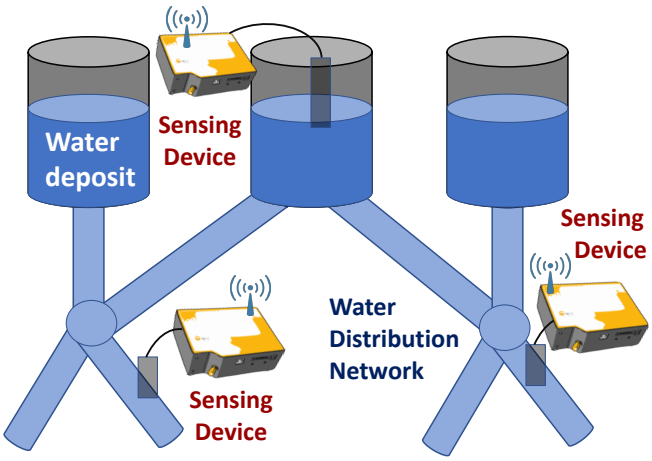

Figure1.1. WP2 workflow

detached or becomes faulty, the network will immediately detect its absence through the expiration of status-reporting messages. More than one device will act as forwarding nodes, to ensure resilience to failures and fault tolerance. Also, water-resistant casing of the hardware equipment and solid placement inside the pilot site infrastructures will guarantee equipment endurance under harsh conditions

A fallback plan can be considered as the following:

a) In case a small percentage of sensor devices produces erroneous results (large percentage of false positives/negatives), then more than one device should be considered to infer whether some pollutant has been detected, or larger quantities of this pollutant (i.e. increase sensitivity).
b) In case a significant portion of sensor devices produces erroneous results, then we should ignore the measurements produced by the sensing infrastructure, until the sensor devices are re-calibrated or until the source of the problem is identified.

The above will increase and ensure the general safety of the sensing infrastructure, as well as the reliability and the reproducibility of the data.

Software tools will be developed to decide on analytical tools that are appropriate for identifying DBPs and materials they interact with. This will involve data fusion of data from conventional methods used for water quality determination (temperature, pH) to estimate DBPs (T2.3). Algorithms and methods based on high-likelihood spatial maps will be applied to wisely choose the locations to monitor the water distribution network. The water distribution utility participant of the consortium (AC) will be modelled and a digital twin will be obtained in order to easily identify the vulnerability of a drinking water utility to contamination threats. This will allow identifying places along the distribution line where the sensing infrastructure should be installed (T2.4). the sensing infrastructure will be deployed to pilot sites of AC (T2.6) and the data will be used to validate the models and algorithms (Figure 1.1).

Methodologies to accomplish O-II A central knowledge base with geographical space and time will be developed to comprise information about DBPs occurrence in drinking water distribution network (T2.5). This will lead to a web-based data geo-visualizations of DBPs and in the first phase will be fed by the information collected by the sensing infrastructure that will be placed at the drinking water distribution network of the AC partner. This will then be opened to any authorized user. The architecture will include two main layers (an upper involving the processors and peripherals and a lower comprising the communication with the analyzer modules). A quality analysis methodology will be used to ensure consistent and standardized implementation. The detailed information (username and password protected) to local operators, regulators and policy-makers while aggregated information (using choropleth maps) will be disponible to the public. This will connect with WP6 aiming for public engagement and awareness to the problems related to water quantity and quality. Moreover, within the framework of WP6 the data collected from WP2 will also allow supporting new policies and regulation regarding water.

Methodologies to accomplish O-III WP3 (Figure 1.2) will gather the data from WP2 regarding DBPs occurrence to understand which are the most hazardous and widespread compounds. Their presence and quantities will be associated with the human health risks at different conditions. Moreover, their fate along time and space will be assessed based on historical observations data (T3.1). Thus, their formation pathways will be assessed as well as their interaction with the materials (naturally occurring and distribution pipes) and other existing pollutants, which will feed T2.3 to allow a more intelligent placement of the sensing infrastructure along the distribution network.

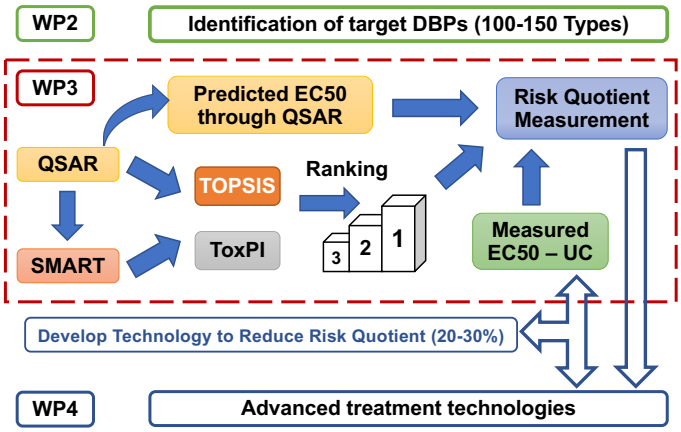

Figure 1.2. Layout of WP3 workflow

Besides regulation gaps regarding drinking water both at the EU level and globally, strategies will be identified to help establishing novel recommendations on prevention measures and policy recommendations (WP6) to protect drinking water whole chain. Several water quality parameters will be modelled using hydroinformatics tools like EPANET and DHI MIKE(T3.5) to help understanding and assess water quality along drinking water distribution network, also considering the possible combined effects resultant from the interaction of other pollutants, through correlations between their concentrations. The models will be validated using data collected from the AC partner distribution network and will consider parameters such as topology, topography, water consumptions, other possible pollutants, and the type of materials, age and possible degradation of the distribution network. The validated model will be sued to predict water quality under several scenarios (extreme events).

Methodologies to accomplish O-IV Indices for drinking water safety plan will be analysed (T3.2) dividing the indicators that are currently included in the standards for water intended to water consumption into 3 groups: indices determining suitability for drinking water (*e.g.* colour, turbidity, iron, *etc.*); indicators that present significant risks to human health (*e.g.* DBPs, hormones, antibiotics) and indicators that pose a toxic threat to human health (*e.g.* Heavy metals, DDT, *etc*.). Other parameters will be included based on the analyses of measures to assess a hazard or threat resulting from extreme events. This methodology will be implemented in water safety plans and to evaluate new strategies in the security management of the water supply systems (T3.4). Moreover, this will be used to assess the efficiency of the water treatment and disinfection technologies developed in WP4. Multi-criteria decision making (MCDM) approaches (Toxicology Priority Index and Technique for Order Preference by Similarity of Ideal Solution) will be used to prioritize target DBPs that can be formed during the disinfection procedure. QSAR models will be used to predict the risk using several end points for the DBPs identified in WP2 (T3.3.). Statistical comparisons will be performed to validate the performance and robustness of the MCDM techniques applied. The risk quotients of the first 20 DBPS ranked as more hazardous will be determined to be used as a reference. Silico QSAR predictions and MCDM techniques will be used to assist in the development of DBPs sensing equipment (WP2) and to evaluate the performance of the water treatment and disinfection technologies from WP4.

Methodologies to accomplish O-V Guidance and practices for preventing water and drinking water contamination will be reviewed bearing in mind data collected in the EU, Israel, USA and Australia (T3.4). This revision will allow to assimilate and propose an extensive list of preventive measures and practices and will focus on those to mitigate the need for intensive disinfection (leading to fewer DBPs formed). A detailed report including advantages and disadvantages/barriers of different options/countries from different perspectives (eco, social and environmental as well as costs and benefits expected) will be prepared. This report will help T6.3 on developing recommendations for EU policy aiming to protect water quality across the whole drinking water chain.

Methodologies to accomplish O-VI Novel advanced and cost-effective drinking water treatment and disinfection technologies will be developed (Figure 1.3). Aiming to replace/reduce the use of chlorination for drinking water disinfection, under-explored features of polychromatic UV will be studied (T4.1). In a first approach, chlorination need will be minimized by combining UV and intermittent chlorination. This approach will be compared in terms of efficiency with standard chlorination. On a second approach, the residual effect of UV will be tested for longer time intervals to verify if UV can totally replace chlorination (one of the advantages of chlorination is its long-term residual effect). The water quality and DBPs formed toxic features will be followed using the tools developed in WP2 and WP3.

Methodologies to accomplish O-VII Novel technologies will be developed (Figure 1.3) to treat water before disinfection (protecting the water sources and removing potential DBPs precursors) and after disinfection (to remove DBPs that were eventually formed during the disinfection step) – T4.2. Organically modified silica (ORMOSIL) aerogels will be prepared in H2OforAll with selected functional surface groups for capturing DBPs and their precursors, *e.g.* amine, thiol and non-polar alkyl groups, which can interact with the DBPs moieties.

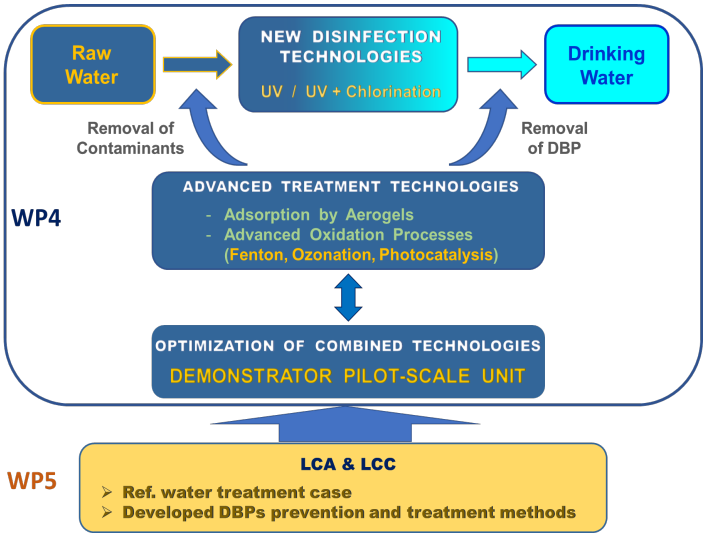

Figure 1.3 Layout of WP4 / WP5 workflow.

Batch and continuous regimes will be implemented for assessing equilibrium and kinetics data, maximizing the removal percentage by controlling the pH of the solutions/wastewaters with DBPs. The use of biochar prepared from several feedstocks (wood, demolished wood, garden waste, sewage sludge, food rest biowaste, cardboard waste) as adsorbent for contaminants removal from water and drinking water will also be considered. To improve their affinity towards the targeted molecules, biochar composites will be developed using physical impregnation and chemical methods, namely by combining them with other porous materials (clay, polymers, and metal-organic frameworks). MOFs possessing high light driven activity will be prepared for the photocatalytic oxidation of the contaminants. Advanced oxidation processes (AOPs) will be studied as suitable technologies for contaminants and DBPs abatement namely: heterogeneous Fenton, photo-Fenton, UV/H_2O_2, photocatalytic oxidation and ozonation. Several catalytic materials will be prepared to enhance the processes efficiency and reduce the amount of chemicals necessary. The technologies will individually be optimized bearing in mind the key operating parameters. Also, the kinetic expressions for degradation will be obtained to have data for the treatment reactors design and scale-up. Moreover, the oxidation processes will be analysed as regeneration methodologies for the loaded adsorbents evaluated in the

same WP. In fact, the adsorbents (aerogels and biochar) may concentrate the pollutants, increasing the degradation kinetics in the oxidation step. The tools developed in WP2 and WP3 will be applied to assess the processes efficiency and help in the selection of the most promising ones.

Methodologies to accomplish O-VIII With the knowledge obtained in WP4 for the individual disinfection and water treatment processes, the most suitable ones for scale-up will be integrated into lab-pilot scale facility that will be installed in drinking water facility in Portugal for demonstration of the efficiency of the integrated processes selected at real conditions of water treatment.

Methodologies to accomplish O-IX A conceptual design of a reference water treatment will be developed including the processes since water supply from the source until drinking water distribution (T5.1). This will serve as a baseline for comparing the performance of the standard water treatment and disinfection procedure with the novel technologies proposed in WP4. Life cycle assessment (LCA) standard tools (ISO 14040-14044) and life cycle costing (LCC) will be followed to assess the environmental performance of the reference water treatment facility (T5.2). Based on the results of WP4, a conceptual design of a novel water treatment and disinfection facility will be developed (T5.3). This model will be used to make recommendations for the new technologies for the best sustainability practice based on LCA and LCC and comparing it with the baseline case (T5.4).

Methodologies to accomplish O-X A web platform for citizens and local communities will be developed (T6.1) and connected to the central knowledge base from WP2 aiming to inform and create awareness regarding water contamination sources within and outside the household as well as recommendation to prevent contamination of water disposed reducing the need for heavy treatments and disinfection (reducing the production of DBPs). The partner AC will test this platform via the open and interactive Water Museum. The outputs from T3.4 will be ranked in a list of recommendations on prevention measures for safety planning. 4 workshops/online consultations will be organized with selected key stakeholders (T6.2) in different locations to provide a review and assessment. Measures for safety planning as well as the main results of the project will be translated to recommendations for policy adaptations at the EU level (T6.3). 3 policy briefs will be drafted and disseminated by the WE partner network and Advocacy Programme for innovative and progressive water policy. The briefs will be disseminated via wider outreach channels and direct communication with policy actors. The last brief will gather the results from the H2OforAll technical WPs to provide practices and recommendations to the Drinking Water Directive and the Zero Pollution Action Plan.

Methodologies to accomplish O-XI Synergies with relevant stakeholders and knowledge integration will be established by organizing one opening dissemination event, participating in third-party events, organization of awareness raising webinars, workshops targeting civil society and a final dissemination event (T7.3). An Exploitation Plan (Figure 1.4) will be developed and implemented (T7.4) including the partner's individual exploitation plans. This will include the organization of 2 events to facilitate the project long term sustainability as well as Water-oriented Living Labs presenting the solutions brought by H2OforAll. The project team will represent H2OforAll WE's exploitation-oriented event (Water Market). Then a second exploitation event will consist of the organization of a focus group to gather conclusions regarding the needs of academia, research organizations, utilities, industry and regulators to assess market gaps and possibilities of long-term exploitation of the project results. T7.5 will ensure the establishment of links with other relevant actions funded under Horizon 2020 and Horizon Europe calls and targeted topics through periodic meetings with sister projects, joint communication materials, a joint policy brief (in collaboration with T6.3) and the organization of a Water Projects Europe event. Development and update of a DCP (T7.1 – Figure 1.4) that will contain the outreach and engagement strategy, communication tools, materials and actions as well as the expected impacts and partners responsibilities including the monitoring and evaluation methodology to be adopted. In T7.2 the dissemination and communication materials and digital tools will be developed to ensure that the project results are disseminated among the relevant actors. This will involve a project website, online dissemination tools (release of articles, posts, educational visual materials, animated video as well as social media profiles) and In-hand dissemination tools (brochures, flyers, roll-ups and other educational materials).

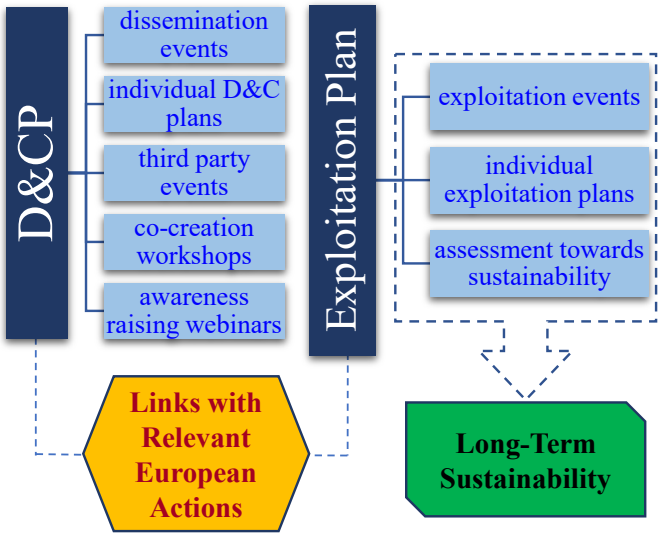

Figure 1.4 Layout of WP7 components.

1.3 Collaboration with national and international research

H2OforAll is aligned with previous and ongoing projects to give a vision on past achievements that will help while looking to the future. Below is a small list of projects in which H2OforAll partners participate: **Sustainable technology for the staged recovery of an agricultural water from high moisture fermentation products,** EU-WATER JPI. Experience making on magnetic char from digestates will be used in H2OforAll. **Development of Catalytic Systems for Effluents Treatment by Photoassisted oxidation**, POCI-01-0247-FEDER-047545. The materials and reactors developed within this project will be applied within H2OforAll framework for DBPs and DBPs precursors removal from water by photocatalytic oxidation. **How to protect water, soil and plants production all together,** EU-Water JPI/006/2016. The relationship between these ecosystems and the know-how acquired with this project involving the protection of water and soil together will be crucial for H2OforAll goals achievement. **MEKOROT –** Environmental surveys for drinking water supply systems for several municipalities. This project will give valuable data regarding drinking water quality within the supply systems. **Cost AERoGELS** – Advanced Engineering and Research of (aero)Gels for Environment and Life Sciences. This Action can support dissemination/mobility stays related to the aerogels' work in H2OforAll. **Aqua2S (H2020)** – Enhancing standardization strategies to integrate innovative technologies for safety and security in existing water networks.

1.4 Interdisciplinarity and integration with social sciences

H2OforAll features a versatile and multidisciplinary consortium with complementary skills regarding analytical techniques development, drinking water management, hydraulics, toxicity analysis, water disinfection, advanced water depollution technologies design, regulation, and citizen engagement. Moreover, it encompasses also important players in the field, such as drinking water management companies, water organizations and technology development (analytical and environmental treatment equipment) stakeholders, ensuring the applicability potential for the targeted technologies. The forthcoming work of the consortium will be focused on the technologies' replicability and scalability (beyond the case-studies analysed within the H2OforAll framework) and their real-scale implementation (helped by the stakeholders). The versatility of the methodologies developed will allow to target other pollutants and ecosystems. The flexibility of the team composed of scientists and stakeholders with the same target *zero-pollution solutions to securing drinking water quality by protecting water sources against pollution, providing innovative monitoring and treatment solutions and ensuring safe distribution* will impact Europe innovation capacity in this field and contribute to a more healthful and prosperous society. To meet the objectives of WP6 and Task 6.1 on citizens' awareness and engagement, we will rely on social science experimental methodologies. Social sciences methods will be applied as follows: (i) to create engaging and effective graphical user interfaces that support the adopted experimental methods. (ii) to assess the level of awareness and engagement by using rigorous experimental methods with control and treatment groups. (iii) to improve the preference elicitation process in the prioritization of the interventions for water quality improvement by relying on insights from social choice theory. An application for ethical approval will be submitted and data will be managed responsibly (GDPR compliance). The engagement of citizens in the study will aim for gender balance and age balance via, for instance, stratifying sampling.

1.5 Gender dimension

The gender dimension is an important consideration in both research scope and the operations of H2OforAll. H2OforAll partners and their organisations are fully aware of the importance of gender issues and are committed to complying with acts, directives and opinions at National and European level (e.g. article 8 of the EC Treaty), Directives 76/207/EEC of 9 February 1976 and the Universal Declaration of Human Rights. In addition, the H2OforAll consortium demonstrates its commitment to promote women in science through direct implementation, as women are expected to lead at least 2 WPs of the project, as well as several tasks within other WPs. The H2OforAll consortium confirms that the partners will take the following actions: i) commit to gender equality in terms of human resources taking into account salary and working conditions; ii) promote an equal opportunity recruitment policy for both genders; iii) the management board will be gender balanced, i.e. max. 2/3 of one gender is allowed; iv) the gender equality aspect of the project will be monitored on a regular basis; v) when selecting stakeholder groups, gender balance and gender representativeness will be pursued; vi) the project aims to maintain a gender balance regarding the experts who will be invited to participate in the workshops, conferences, seminars, etc.

1.6 Open science

Data management plan (WP7) will be the central point for data handling. This will follow the FAIR principles, to ensure that researchers can find, access and re-use data. All scientific publications of this project will be open access, predominately golden open access following the Open Access policy. A H2OforAll ZENODO community will be created where it will be uploaded the bibliographic metadata and related research data linked to the publications as well as other dissemination material such as presentations and posters. At the same time, when articles containing the project data will be published, the article will refer to the data by its DOI and contribute to the success of the project during and beyond the funding time. Articles publication will be leveraged by short webinars to be made available online summarizing the importance of the results. These webinars will be distributed via H2OforAll and

the Consortium partner's network. Edited records of webinars will be grouped and made available on YouTube.

1.7 Data management

The actual data management plan (DMP) will be written in the first 3 months after the funding of the project, following the data management policies of every partner. This will be done in a written agreement aiming to share as much data as possible. The DMP will be regularly reviewed and updated to ensure that the data management is consistent throughout the project lifetime. During the project kick-off meeting, also a data-management expert group will be established. This will be responsible to create and maintain the DMP. To create a DMP the HORIZON EUROPE template will be used. Using such a DMP source enables to share the data management plan with all co-owners, as it provides access to the DMP to every consortium partner. The DMP will describe the type of data produced and collected, the owner of the data and the access rights and modalities, how data will be cured and preserved during the project duration and how it will be handled afterwards. FAIR principles will be applied. This will be ensured by adding the data to national data repositories and at an international repository like Zenodo. By adding the data to such repositories, the data get a DOI, a meta data description gets created, including clear licensing by creative common licenses (cc). Both Zenodo and national repositories are trusted repositories. The DBPs monitoring around Europe will be made available to drinking water operators and regulators. Moreover, an interface with summarized information (to protect local operators' sensitive information) will be made fully available to the public. These data will also be shared via the IPCHEM platform. Adding data will be done at the time of submitting a publication the latest. Additionally, information on the research design for each scientific research will be added to the data so that research reproducibility can be provided. To ensure that the research data are secure and preserved the data will also be saved in the institutional archives. Personal data collected in the project (*e.g.* in engagement activities in WP6) will be treated carefully and confidentially and will not be shared without the prior written consent of the data subjects, in line with the EU Data Protection Directive. Data will be kept available at least 5 years after the project ends. The ownership and commercial rights of the results generated within the project will be detailed and agreed upon the Consortium Agreement-CA (DESCA model). Also, all partners will sign a Non-Disclosure Agreement to guarantee the confidentiality of the results and will observed the European Charter for Researchers and Commission Recommendation on the management of the intellectual property. The results considered as potentially patentable will be made fully available as soon as the patent application is submitted.

2. Impact

2.1 Project's pathways towards impact

The Section below shows how the proposed activities of H2OforAll address the expected outcomes (EO) and impacts (EI) of the topic HORIZON-CL6-2022-ZEROPOLLUTION-01-04.

EO1. A wider use of a better understanding and an enhanced knowledge base is required to assess pollution sources, pathways and combined effects on drinking water systems, including forward looking approaches aimed to anticipate and prepare for future or emerging challenges. A better understanding of DBPs formation, and transmission pathways will allow to stablish measures to prevent/minimize their production. The interaction of the contaminants present in raw water with the disinfectant will be unravelled aiming to establish preventive measures to avoid DBPs formation. The knowledge of DBPs behaviour along space and time as well as their interaction with natural occurring materials and pipelines will allow predicting with high precision where, along the distribution network, DBPs monitoring infrastructure shall be introduced. This will allow an intelligent decision of where and how disinfectants reinjection will be required along the pipeline from the source to the consumers reducing the need for disinfection. **Target groups:** Drinking water management companies; consumers; scientific community

EO2. Implement advanced preventive and mitigating strategies and measures to protect drinking water sources, treatment and supply against harmful effects of global and climate change. H2OforAll research will provide a wide range of preventive measures to protect drinking water that will be built based on the regulations and practices of different countries (EU, USA, Israel and Australia) giving a guideline for practices that should be applied by stakeholders and private households in order to avoid the need of intense disinfection. The preventive measures will bear in mind emerging challenges regarding climate change and extreme events by establishing strategies that allow to protect the water sources and supply even from extreme events. **Target groups:** Drinking water management companies; consumers; Industrial stakeholders; Policy-makers

EO3. Apply effective risk assessment and risk management strategies enabling early warning systems and delivering ready-made outputs for decision-making and governance. MCDM will allow to prioritize the main DBPs based on their risk. Moreover, the new disinfection technologies will lead to different DBPs whose ecological and human health will be predicted based on silico QSAR and MCDM. The data and tools generated will be key for assessing the quality of novel disinfection strategies as well as helping on the design of novel sensing systems

enabling suitable monitoring of the drinking water chain for early warning systems and supporting decision-making and new regulation. **Target groups:** Drinking water management companies; Policy-makers; Scientific community

EO4. Exploit advanced, integrated and cost-effective water quality sensors and analytical methods. H2OforAll targets the development of high-resolution sensors able to promptly determine and quantify targeted DBPs in drinking water. We aim to further develop (after the project lifespan) the prototype that will be reached. Thus, robust, validated, and portable analytical tools will be available to follow DBPs concentration. The project will also give the scientific basis for further analytical technologies development. Depending on the costs of the sensors to be developed, this may be made available to the public so that the consumers can assess the quality of the water there are receiving. **Target groups:** Drinking water management companies; Scientific community; public

EO5. Deploy innovative and robust monitoring systems and real-time information on drinking water quality, from sources to supply. The novel sensors for DBPs detection and quantification that will be developed will be part of a robust monitoring structure. Within H2OforAll framework, algorithms will be developed (based on the knowledge of DBPs behaviour along time and interaction with materials) to wisely choose the locations for monitoring the water distribution network. To be efficient, this monitoring will target the key DBPs that will be prioritized based on their potential risk. This innovative infrastructure will give real time information on drinking water quality from sources to supply. **Target groups:** Drinking water management companies; Scientific community; policy-makers; public

EO6. Disseminate and use a robust knowledge on the occurrence, persistence and degradability of disinfection by-products (DBPs) in drinking water with due consideration to operational parameters, chemicals, materials and biofilms interactions, including the pathways related to human exposure A CKB will be developed where geographical space and time information about DBPs will be feed by authorized users. It will be available to regulators, policy-makers and drinking water infrastructure operators. It will have an interface for the public to raise their awareness to the occurrence, persistence and degradability of DBPs giving a big picture of their distribution in several EU countries. The risk assessment studies to be performed will allow identifying the DBPs more likely to impact on human health. The public engagement will create awareness for the pathways related to human exposure to DBPs and potential consequences. **Target groups:** Drinking water management companies; Scientific community; policy-makers; general public

EO7. Spread the use of advanced and cost-effective drinking water treatment and disinfection processes and technologies, including transformative approaches Alternative drinking water technologies will be targeted in H2OforAll involving unravelled potential of MP UV radiation. This will reduce the need for chlorination leading to lower levels of DBPs in drinking water. Moreover, innovative treatment approaches involving adsorption and advanced oxidation using novel and costs effective materials will be developed to remove contaminants from water before disinfection (to abate potential DBPs precursors such as algal blooms precursors) and/or after the disinfection stage (to remove DBPs that were formed during that stage). Environmental and economic sustainability of the proposed processes will be compared with the reference water treatment case to ensure their cost-effectiveness. The data that will be obtained will impact in water treatment science and technology with potential for commercial novel processes that will lead to a safer drinking water. **Target groups:** Drinking water management companies; Scientific community; policy-makers; public

EO8. Broad uptake of advanced knowledge, breakthrough solutions and innovative technologies to enhance the competitiveness of the EU water sector and fostering the EU's position and role in the global water scene. H2OforAll will provide advanced knowledge, breakthrough solutions and innovative technologies for the water sector at several levels: - by developing advanced sensing solutions in a breakthrough real-time monitoring infrastructure. By developing innovative methodologies to prioritize DBPs. By unravelling the DBPs formation mechanism and interaction with materials along the drinking water distribution network. By developing breakthrough technologies able to disinfect water without the formation (or with lower concentrations) of DBPs and innovative procedures to remove DPBs precursors and DBPs from drinking water. By protecting the water sources from pollution through prevention as well as by developing raw water treatment technologies. The results to be obtained will be the scientific and technological basis to make the EU a leader in the global water sector **Target groups:** Drinking water management companies; Scientific community; policy-makers; general public

EO9. Increasing the EU scientific and technological base and guidance on measures to manage drinking water quality and evidence for policy-making, safety planning and implementation. H2OforAll consortium is committed to the FAIR (Findable, Accessible, Interoperable, and Reusable) data approach. Thus, bearing in mind IPR management, the results obtained in this project will be made accessible and comprehensive to the public. H2OforAll will also have a strong impact on supporting policy development, regulatory action and communication at different levels based on the scientific and technological developments that will be achieved within the framework of the project. All data (DBPs monitoring, behaviour, prevention and treatment) will give support to future legislation and actions to protect drinking water. Moreover, H2OforAll will provide suitable cost-effective assembled solutions for water treatment and disinfection with the lower formation of DBPs, giving a key tool to the stakeholders to fulfil the

future tighter environmental legislation. The respective dissemination activities (including online events addressing the broad policy circles – decision-makers, monitoring bodies and authorities) will increase the visibility of the above outputs and so maximize the support to policy-making. Dedicated policy recommendations produced in WP6 will gather key findings to policy makers from H2OforAll WPs. **Target groups:** Drinking water management companies; Scientific community; policy-makers; public; Stakeholders

EO10. Science and evidence-based implementation of the European Green Deal and the Sustainable Development Goals, notably the SDG 6 "Ensure availability and sustainable management of water and sanitation for all". H2OforAll is totally aligned with the SDG6 as follows: Development of prevention strategies for protecting drinking water that will help educate stakeholders and the public about the importance of protecting drinking water from sources to supply also involving wastewater management and disposal (SGD target 6.1). This public engagement will also target education regarding appropriate water behaviour by enrolling the global water quality and scarcity issues related with consuming habits in order to reduce water waste and contamination. The innovative disinfection and water treatment technologies using sustainable materials will allow proper water treatment (source) and disinfection without the formation of noxious DBP (or limiting their production) – SGD targets 6.3 and 6.6. The technologies to be developed may be optimized to be more cost-effective and made available to lower-income countries contributing to ensure available and sustainable management of water and sanitation for all. The development of suitable DBPs sensing strategies within a suitable monitoring infrastructure will allow water management companies, regulators and consumers to follow up the quality of their water and putting in action the preventive measures that will be stablished within H2OforAll whenever necessary (SGD target 6.5). The strong public engagement and policy recommendations that will be made available within the framework of the project as well as preventive measures that will be stablished and novel disinfection and water treatment strategies will push policy-makers and regulators to establish harsher drinking water protection regulations to ensure availability and sustainable management of safe water for all (SGD targets 6.1 and 6.b). **Target groups:** Drinking water management companies; Scientific community; policy-makers; public; regulators

EI1. Advanced understanding of diffuse and point sources of water pollution in a global and climate change context, enabling novel solutions to protect water bodies, aquatic ecosystems and soil functionality, and further enhancing water quality and its management for safe human and ecological use, while fostering the EU's and Associated Countries' position and role in the global water scene. A better understanding of DBPs formation pathways that will be reach through H2OforAll will give the necessary tools to understand the role of water pollution on DBPs formation as well as to predict the behaviour of water treatment and disinfection technologies efficiency as function of the source water quality. The preventive measures that will be stablished in WP3 aiming to protect drinking water quality whole chain will bear in mind also the protection of the water bodies (sources of drinking water). Besides, advanced water treatment technologies aim to remediate water from the source by reducing its pollution load leading thus to better aquatic ecosystems and enhancing water quality and its management for safe human and ecological use. **Target groups:** Drinking water management companies; Scientific community; policy-makers; public; regulators

2.1.1 Requirements and potential barriers

Some barriers and limitations may arise that may hinder the complete fulfilment of expected impacts (Table 2.1). This consortium of 18 partners from all regions of Europe will gather synergies and do whatever possible, during the project time period and beyond that, to mitigate all the shortcomings that may appear. Moreover, future projects will certainly be proposed to follow-up the work and face these challenges.

Table 2.1 Possible barriers/obstacles to achieve the expected impacts

Type	Barrier
Regulation	Regulation change is a time-consuming process. The new data gathered by H2OforAll that will allow identifying target DBPs and their toxic effect as well as their "map of contamination" will surely impact decision-making, but it will take time for a new related regulation to enter into force. The preventive measures that will be proposed will imply habit and procedures changes. There is always some inertia and fear towards change. The consortium aims, within the framework of the project and beyond it, to engage the public, stakeholders and policy-makers on this problematic to overcome such barriers.
Standards	New standards for source water and drinking water quality will be proposed based on the H2OforAll results. The adopting of such standards by the drinking water management companies may be a barrier. We aim to provoke the public and policy pressure within the project (WP6) and beyond it through follow up actions such as workshops, press-releases and from results communication to overcome the barrier.
Public acceptance	H2OforAll will require the public acceptance to change the consumption pattern to avoid water pollution and waste. This change may be difficult since it will imply correcting "old" habits. The public may fear the safety of the drinking water coming from the novel disinfection and treatment technologies

	proposed. Also, stakeholders' acceptance of new water treatment and disinfection technologies is challenging. The public engagement strategies by disseminating the science in an understandable way and by targeting mainly the young generation will surely impact in habits changing. The novel technologies will be validated both in efficiency and LCA/LCC. The data will be shared among stakeholders and the public to motivate their acceptance. The legislation and regulation change we aim to provoke will also promote the acceptance.
Workforce considerations	Specialized workers will be required to implement and operate at the industrial scale the novel monitoring infrastructure and water disinfection/treatment technologies. This may be a barrier to the implementation of H2OforAll technology if the industrial stakeholders are not willing to invest. On the other hand, if well accepted, it gives new job opportunities to qualified human resources. Since we have in the consortium companies that develop the technologies (Adventech, IRIS and Atlantium) as well as end-users (AC), their action in the field will overcome such barrier.
Financing of follow-up steps	H2OforAll will lead to validated integrated assembled technologies for DBPs monitoring and removal from drinking water as well as novel disinfection strategies. At the end of the project validated lab-pilot units will be reached. However, further investment will be required after the project to obtain a demonstrative unit at a larger scale to demonstrate the efficiency at the stakeholders' premises. Thus, H2OforAll consortium will further seek for new financial incentives both at the national and European level.
Cooperation of other links in the value chain	H2OforAll success will require the involvement and cooperation with stakeholders' from water treatment companies, drinking water distribution stakeholders, consumers as well as decision-makers. This cooperation will be facilitated with the involvement of main players as partners in the consortium as well as in our advisory board.

2.1.2 Scale and significance of the project's contribution to the expected outcomes and impacts

H2OforAll results will contribute to the call outcomes and are described in scientific, societal and economic impacts that will be the fostered by dissemination and exploitation measures described in section 2.2.

Scientific impacts: 80% of the technologies that will be studied in H2OforAll (involving sensing systems, monitoring infrastructures, drinking water disinfection and water treatment) are based on the high risk – high gain principle. The remaining 20% (*e.g.* toxicity assessment) are based on satellite work, leveraging the potential of the technologies to demonstration level. The success of the project will create and diffuse (through WP7) high-quality new knowledge in the field of analytical techniques and devices; intelligent monitoring systems and water treatment and disinfection using innovative and integrated technologies. Moreover, new algorithms will be stablished for selecting monitoring spots by grasping DBPs behaviour and interaction with materials along the drinking water system. This will allow reaching prediction models able to simulate DBPs fate along the network as well as which are the most likely DBPs formed when different disinfection technologies are applied. Also, new knowledge and tools will be developed by integrating MCDM techniques that will allow to identify the most problematic compounds as well as predicting the risk of new molecules through in silico QSAR predictions. Scientific innovation will render 20 peer reviewed scientific publications as well as participation in 40 international conferences. H2OforAll strengthens human capital at the individual and consortium levels through scientific training in the form of 8 PhD and 18 MSc in the framework of the project. Some supervision/mentoring activities will be shared between H2OforAll partners, this will allow young researchers career development by engaging them in a multicultural and multidisciplinary research environment. The know-how generated in H2OforAll regarding the DBP behaviour and fate as well as the efficiency and design of novel disinfection and water treatment technologies will be included in 4 curricular units at pre- and post-graduation levels. Besides, a technological facility at the lab-pilot scale will be made available to further tests at the relevant environment (monitoring infrastructure prototype and integrated disinfection and water treatment facilities) making a bridge between the academia and stakeholders. After H2OforAll, the impact on human capital at the scientific level will be assessed as the number of trained researchers with impact on the scientific field they develop in H2OforAll or new lines of research created after knowledge integration.

Societal impacts: H2OforAll strengthens the uptake of innovation in society. H2OforAll will develop a CKB with updated information about DBPs through the visualisation of their occurrence in several countries. This will be an interface for operators and regulators to get real-time information related to DBPs. Moreover, an interface for the public aggregating summarized information will be available. H2OforAll fosters the diffusion of knowledge and open science by sharing all public research outputs (i.e. 70% of total: not protected by IP) through open knowledge infrastructures. Care will be taken to make science and technology understandable by public (see section 2.2). Also, the preventive measures to protect drinking water will target water distributors, stakeholders and consumers. This will raise societal awareness on DBPs issues which will have an impact as a driving force on policy-making and regulations. H2OforAll addresses EU priorities in line with the European Green Deal's zero pollution ambition by

enabling novel solutions to protect water bodies and further enhancing water quality and its management for safe human and ecological use. H2OforAll will also contribute to increase the quality of living of 68% of the world's growing population estimated to live in urban areas by 2050 by promoting a higher quality of drinking water supplied (reducing the DBPs levels below WHO guideline values) and by preventing water sources contaminations. The reliable monitoring system and cost-effective treatment and disinfection technologies to be developed as well as the preventive measures to be widely spread will help to ensure availability and sustainable management of water for all (in line with SDG6). H2Oforall expects that at least 50% of innovation and scientific results targeting EU goals are shared and integrated thus contributing to policy-making through norms, standards and regulation (WP6).

Economic impacts: H2OforAll foresees 4 potential innovative technologies/patents and IPR (based on the novel sensing and monitoring system as well as advanced disinfection and water treatment technologies to be developed). All submitted applications will be followed in the medium term and the number of awarded (*e.g.* Patents) will define the impact of H2OforAll, as well as the market shares that developed the innovations (*e.g.* water treatment companies). The implementation of novel raw water treatments able to remove DBPs and algal blooms precursors will reduce the drinking water disinfection costs by at least 5%.

2.2 Measures to maximise impact - Dissemination, exploitation and communication

2.2.1 Dissemination and Communication Plan

The **Dissemination and Communication Plan,** will be developed under WP7, a cross-cutting WP that will bridge all the project WPs and foster aligned engagement and long-term impact of the project activities and results.

Objectives of the Dissemination and Communication Plan

The D&C Plan will include the overall strategy, objectives and expected outcomes, target audiences, outputs for dissemination, dissemination materials, tools, and channels, as well as KPIs. The D&C Plan will disseminate those key messages, including the progress of the project activities and materials (such as publications, educational resources and one educational kit) while engaging the main target groups (such as civil society, academia, research organisations, utilities organisations, industry and regulators). The D&C Plan will cover the entire duration of the project and will result in online and in person activities which will attract the defined target audiences, raise awareness on DPBs in drinking water and engage stakeholders in various project activities that will be led through a co-creation lens whenever possible. The dissemination strategy will make use of the EC projects' communication best practices and follow the 5W+1H approach to disseminate and communicate: **Why:** to promote and disseminate the objectives and progress of the project activities. **What:** key messages and information about DPBs in drinking water, guidelines and knowledge for managing drinking water quality, project's goals, outcomes, publications, and community engagement activities. **To Whom:** to the main target groups of stakeholders: civil society, academia, research organisations, utility organisations, industry and regulators. **Where:** through the networks of project partners, the target stakeholders, as well as the project website, newsletters, social media, events, and promotional materials. **When:** starting from the launch of the project and adapted to the timeline of the project; **How:** a tailored approach to the specific needs of the target groups, mostly through online and in-print.

Target Groups

The D&C Plan indicates the activities that will be undertaken by the partners throughout the project lifetime, outlining the target groups, guidelines and communication channels to be used in order to ensure the effective implementation of the strategy. H2OforAll envisions active stakeholder engagement for wider knowledge dissemination amongst the public and those involved in ensuring water safety (Table 2.2).

Table 2.2 Stakeholders engagement for targeted knowledge dissemination

Target group	Definition	Goal
Scientific and technology community	Higher education institutions, research labs and centres, science journalists, and research networks connected to water issues	To disseminate the project scientific and research outputs, conclusions amongst the R&D community, and develop scientific and research results and analysis, enabling the scientific community to leverage the activities and results developed
Water utilities organisations and industry	Water utility companies, water treatment companies, water supplier companies, water operators, other large industry and SMEs that are part of water treatment value chain	To broaden dissemination and exploitation opportunities and assessing the potential of the solutions and technologies from the project, contributing to the development of new R&I outputs and products to the market
Policy makers, regulators,	National & European policy makers (municipalities, inter-governmental bodies, EU agencies, *etc.*) related to water safety policies and regulation	To disseminate transformative prevention and mitigation approaches to DPBs in drinking water at the legislative level, and showcase water treatment solutions for policy intervention

governmental entities		
Civil society	Community groups, NGOs, professional associations, foundations and citizens	To promote horizontal knowledge dissemination, engage civil society and promote information through the wider public
Media	Mass media, online news outlets	To make information widely accessible and easy to reach by citizens

Communication channels and tools

To reach the desired target groups, the following communication activities (Table 2.3) will be included in the dissemination plan. The communication activities will be developed in WP7 with the collaboration of all partners.

Table 2.3 Communication channels and activities

Channels and tools	Description	Target Groups	Month
Project website	A website dedicated to the project will be launched and updated with news, publications and educational materials throughout the duration of the project	Research organisations; Civil society	M3
In-hand and digital dissemination tools	Project brochures, flyers, infographics, a roll-up, and one educational hands-on kit consisting of materials related to the project research to promote the project online and in-person events. The materials will be developed in a digital format and printed if needed.	Research organisations; Water utilities, industry; Policy makers, regulators, governmental entities; Civil Society	M6, M32
Social media	Accounts on LinkedIn, Twitter, Facebook, Instagram		M2
E-Newsletter	Every 3 months, an e-newsletter providing extensive information on the progress of the project will be managed through Mailblast, and a downloadable PDF version will be available on the project website		M6, M12, M18, M24, M30, M36
Press releases	Press releases will be issued for specific project activities as well as at the end of the project in order to detail its main achievements and the overall impact	All	M1-M36
Publications (Scientific)	Scientific publications will be submitted to academic journals	Research organisations	M12, M24, M36
Publications (Educational)	Online educational materials will be developed in order to ensure wide dissemination of the project research amongst the general public and the relevant stakeholders.	Civil Society	M6, M12, M18, M24, M30, M36
Videos	Videos detailing the main conclusions of the project.	All	M6, M18, M36
Events	**Opening event:** the consortia will introduce the project team, objectives, and methodology.		M2
	Closing event: the consortia will disseminate the main conclusions and exploitable results of the project.		M36
	Third-party events: the consortia will join third party events when relevant for the dissemination of the project results(*e.g.* WaterJPI; WaTech; *Encontro de Entidades Gestoras de Água e Saneamento*; Water Innovation Europe, Water Projects Europe).		When relevant
	Awareness raising webinar series: 6 webinars focusing on relevant thematic areas related to safe drinking water whose topics will be defined by the project coordinator and inspired by the WP's progress and related research.		M6, M12, M18, M24, M30, M36
	Co-Creation Workshops: 2 workshops (Civil Society; Research organisations) and (Water utilities organisations, industry; Policy makers, regulators, governmental entities) for the dissemination of knowledge and of prevention and mitigation practices related to DBPs.	**Workshop 1:** **Workshop 2:**	M18, M32

A project internal Mailing List will be developed to ensure that all project staff members are included in all interactions. Teleconference Meetings with the project partners will be organised regularly (via *e.g.* Skype, Zoom) to discuss the progress. Additional meetings with other relevant partnerships and networks will be organised as needed to empower the dissemination and communication actions. In this context, the selected metrics and target Key Performance Indicators (KPI) for dissemination and communication activities are specified in Table 2.4. These will be regularly checked by the means of Google Analytics, metrics from the website's back-end, partners' reporting actions and listing of all social media posts, press releases, e-newsletters, and other digital content developed.

Table 2.4 Communication metrics and KPIs

Channels and tools	Metric	Target
Project website	No. of visits	10,000 (total)
In-paper materials	No. of printed brochures, flyers, infographics, educational hands-on kit distributed	800 (total)
Digitally distributed materials	No. of educational materials developed No. of downloads	6 (total) 300 (total)
Social media pages	No. of followers No. of posts	1,000 (total, across all pages) 90 posts (total, across all pages)
E-Newsletter	No. of subscribers No. of newsletters sent	At least 250 6 (total)
Press releases	No. of publications in the project website	36 (total)
Publications	No. of scientific publications	20 published
Events	No. of dissemination events (opening, closing, awareness raising webinar series, co-creation workshops, focus group) / No. of participants	11 events (total) / Opening event (80), Closing event (80), Webinars (15 each), Co-creation workshops (10 each), Focus group (15)
	No. of third-party events attended	10 (total)
	No. of conferences attended	40 (total)

It is also relevant to note that each partner will leverage its individual dissemination networks and channels to further promote the activities developed under the project. In particular, the parts will use the following channels and tools to sustain the communication impact of the project.

Communication and dissemination through the official organisation's accounts: the consortium will utilise its social media accounts and website coverage to further promote H2OforAll. In particular, the consortium will make use of its social media channels, which include over 200,000 social media followers.

Leveraging the partners' networks for communicating H2OforAll: the consortium will target specifically the networks of water organisations that are part of the consortium as partners, namely Water Europe and Wetsus. Besides this, other networks will also be targeted, such as: AOPs PhD School, TWI Innovation Network, European University of the Seas - SEA-EU, ScanBalt BioRegion, Marine Biodiversity and Ecosystem Functioning, EU Water JPI, SPIRE, WaterCampus, WaterAlliance, ERRIN, Water4all Partnership, EIT Raw Materials and Netherlands Water Platform, Parceria Portuguesa para a Água, Associação Portuguesa de Distribuição e Drenagem de Águas, SECPHO-LIGHT Technologies CLUSTER, European Photonic Industry Consortium, and water distribution utilities and waterboards around the Netherlands including Vechtstromen, Vitens and WMD Drinkwater.

2.2.2 Exploitation Plan

The **Exploitation Plan** will define a set of Key Exploitation Results (KERs) – Table 2.5, strategies to address barriers related to maximizing impact, Memorandums of Understanding (MoUs), individual exploitation plans, organisation of a focus group for project exploitation and a business plan. The Consortium foresees the main KERs will result from **WP2: DBPs Sensing, Analysis and Visualisations, WP3: Environmental Impact and Risk Assessment of DBPs, WP4: Advanced and cost-effective technologies for the treatment of drinking water,** and **WP6: Guidance on measures, policy and public engagement.**

The consortium is committed to making the data available in open formats during the project and free of charge for the EC and all stakeholders to use and publish, along with other relevant reports tailored for the use of the EC. In the post-project period, one of the key challenges will be balancing this with the need to develop revenue streams to finance the maintenance and further development of the DPBs treatment solutions and processes.

Table 2.5 KERs description and target groups

Key Exploitable Result		Description	Target Groups
KER1	Patents	Sensors and sensing infrastructures for the identification of DBPs **(T2.1, T2.2, T2.3, T2.4)**	Municipal and private water utilities organisations, water treatment and supplier companies, operators and regulators; National & European policy makers particularly those related to water safety policies
		Cost-effective treatment technologies for drinking water **(T4.2.5)**	Water utilities organisations, water treatment and supplier companies, water operators and regulators; National & European policy makers particularly those related to water safety policies; Community groups, NGOs, professional associations, foundations and citizens
KER2	Open access tools	Real-time interactive maps of DBPs from sources to supply **(T2.5)**	Higher education institutions, research organisations and research networks connected to water issues; Municipal and private water utilities organisations, water treatment and supplier companies, water operators and regulators; National & European policy makers particularly those related to water safety policies; Community groups, NGOs, professional associations, foundations and citizens.
		CKB with information about DBPs **(T2.5)**	
KER4	Processes	Disinfection processes to prevent/minimize DBPs production **(T4.1)**	Municipal and private water utilities organisations, water treatment and supplier companies, operators and regulators; National & European policy makers particularly those related to water safety policies
KER5	Guidelines	Recommendations on Prevention Measures for Safety Planning **(T6.2)** Indices for drinking water safety plan **(T3.2)**	Higher education institutions, research organisations and research networks connected to water issues; Municipal and private water utilities organisations, water treatment and supplier companies, water operators and regulators; National & European policy makers particularly those related to water safety policies; Community groups, NGOs, professional associations, foundations and citizens. Online news outlets

Furthermore, within the Exploitation Plan, partners will define and develop individual exploitation plans.

UC	UC will use to disseminate the collected information to the public by open access publications. UC intends to apply for follow up project calls to develop the drinking water disinfection and treatment technologies to reach higher TRLs targeting their commercialisation (aided by the companies and/or eventual spin-offs). Cutting-edge achievements will be included in MSc course topics. Training of MSc and PhD.
UULM	UULM intends to pursue the development of the optical sensing devices for water monitoring to higher TRL levels so that such structure will be commercially available.
IRIS	IRIS will be using the findings and results prior and within the H20forAll for market deployment. Will be keen on possible licensing and/or patentability of the developments made within the tasks carried out.
ULeic	ULeic will use the key findings on the MCMD approach and the experimental results from the work packages in H2OforAll to present in international conference, and publish in open access international journals, figshare/metadata repositories.
CYENS	CYENS will inform the water boards of Cyprus on treatment technology opportunities as well as new ways for detecting pollutants by way of the project's results.
UHaifa	UHaifa will publish the research results in at least 2 peer review journals and include the results in a master student thesis. The findings will also be disseminated in at least 4 national/international conferences.
UGdansk	UGdansk will use the results of the project for further research. Promising data will be disseminated in publications as well as presented at the conferences.
KTH	KTH will write at least 2 scientific journal papers and use the results for the education of 50 master students/year at KTH. Project results will be part of KTH PhD thesis; inform the Swedish Water Committee through the KTH Water Centre; and contribute to the scientific conference and workshop EU Water JPI.
UNIVLEEDS	Will pursue the public engagement and awareness by maintaining and updating the web-platform.
Adven-tech	Adventech is planning to uptake the technologies beyond the project and to provide larger scale demonstrations to reach market readiness.

UT	UT is collaborating with various water distribution utilities and waterboards around the Netherlands. These bodies are striving for new technologies that provide better sensing possibilities for real-time detection of pollutants, and they are very interested to invest on these technologies, using UT as a potential partner. Results of the project would be part of the curriculum of various courses of the university and will create new opportunities for research and PhD fellowships.
WE	WE as the European Technology Platform for Water and leveraging from its extensive network (over 18,000 readers) will promote H2OforAll in its conferences as well as in all its media channels. WE will promote H2OforAll in its new series of dedicated events (Water Project Europe) to gather and capitalise the experiences of compatible projects for policy, collaboration and further development of the TRL and will collaborate with existing Water oriented Living Labs for the dissemination and exploitation of results.
Atlan-tium	Atlantium will communicate the results to their global representatives and costumers, and their marketing department which will distribute it on various channels, including direct communication of R&D with company representatives and big clients in US, China, and Southeast Asia. Patent applications are foreseen.
IVL	IVL will share project results with the scientific community by means of scientific publications, and with relevant public bodies and industries through applied consultancy projects in the water supply sector.
AC	AC will present project results in water sector conferences and meetings of stakeholders. In addition, "Museu da Água" is a platform used to reach all types of public and generations, through a diversified set of initiatives, seeking to transform citizens into active participants in the protection and defence of water.
Wetsus	Wetsus has a broad network of water technology partners, both companies (>100) and knowledge institutes (>20) and will focus on capacity-building and forging partnerships, including collaboration with external stakeholder-partners to ensure early adopter's uptake of the project results. This includes patenting.
SPI	SPI will use the knowledge, expertise, and network developed during the previous and current Horizon Europe and H2020 projects to widen H2OforAll's exploitation and market deployment.

2.2.3. Strategy for IP management

The project will adopt a **policy of protection of the project's results** that are suited to be commercially exploitable and to be protected. Strategies for innovation and IPR management will be defined within WP7 and integrated in the project's final exploitation plan. A Consortium Agreement will be signed at project start and will act as the legal framework. To rightfully determine the ownership of results, the following approach will be adopted: Definition of pre-existing knowledge of the partners (Background). The Consortium Agreement will state each background included and excluded from Access Rights; Mapping of existing patents and potentially overlapping IPR; Assessment of the new knowledge to be generated during the project, including the creation of an IP directory to determine which partners are the owners of what IP. During the project duration and beyond (several years, to be agreed in the CA), the contractors shall treat any information that is designated as property by the disclosing contractors as confidential. They shall also impose the same obligations on their employees and suppliers. The ownership of the knowledge developed within the project will be governed by an open-source license and approach.

2.3 Summary

KEY ELEMENT OF THE IMPACT SECTION

SPECIFIC NEEDS	EXPECTED RESULTS	D & E & C MEASURES
What are the specific needs that triggered this project? Need to mitigate the low quality of water sources due to contamination of soil and freshwater Need to foster the protection of the drinking water chain in regard to DBPs produced when chlorine disinfectant reacts with organic matter in water Need to expand a limited knowledge on the toxicity and impact of DBPs on the environment and human health Need to mitigate the impacts of pathogen development in water sources and lake eutrophication (linked to increased algal blooms) resulting from climate change Need to build society and governments' awareness of drinking water contaminants.	What do you expect to generate by the end of the project? **KER1**: Sensors and sensing infrastructures for the identification of DBPs and/or the main materials they interact with **(TG2, TG3)** **KER1:** Cost-effective treatment technologies for drinking water **(TG2, TG3, TG4)** **KER2:** Real-time interactive maps of DBPs from sources to supply **(TG1, TG2, TG3, TG4)** **KER2**: Central Knowledge Base where information about DBPs can be added worldwide, associating this information with space and time, allowing the development of web-based data geo-visualisation **(TG1, TG2, TG3, TG4)** **KER3**: Advanced and cost-effective alternative disinfection processes to prevent/minimize DBPs production **(TG2, TG3)** **KER4**: Recommendations on Prevention Measures for Safety Planning **(TG1, TG2, TG3, TG4, TG5)** **KER4**: Indices for drinking water safety plan **(TG1, TG2, TG3, TG4, TG5)**	What dissemination, exploitation and communication measures will you apply to the results? Dissemination: Via one opening dissemination events organization of an awareness raising webinar series, development of two-creation workshops, organization of one final dissemination event, development of partner's individual dissemination plans, participation in 10 third party events and 40 conferences. Exploitation: Via definition of key exploitable results (KERs), development of strategy for intellectual property rights (IPR) management, cooperation with water-oriented Living Labs (WoLLs), participation in Water Europe's exploitation-oriented events, organisation of one focus group to assess market gaps and possibilities for long-term exploitation of the project's final results, and 20 open-source scientific publications Communication: Via project website, in-hand dissemination tools (such as brochures, flyers, roll-ups and other educational materials), online dissemination tools (such as e-newsletter, educational visual materials, videos, press releases) and seeking complementarities and synergies with other relevant EU-funded projects and actions

TARGET GROUPS

Who will use or further up-take the results of the project? Who will benefit from the results of the project?

TG1: Higher education institutions, research organisations and research networks connected to water issues;

TG2: Municipal and private water utility organisations, water treatment companies, water supplier companies, water operators and water regulators;

TG3: National & European policy makers (municipalities, inter-governmental bodies, EU agencies, *etc.*), particularly those related to water safety policies;

TG4: Community groups, NGOs, professional associations, foundations and citizens;

TG5: Mass media, online news outlets

OUTCOMES

What change do you expect to see after successful dissemination and exploitation of project results to the target group(s)?

Better understanding surrounding DBPs formation and transmission pathways which will allow the establishment of adequate measures to prevent/minimize their production, as well as mitigate their impacts (**EO1, EO2)**

New disinfection technologies will lead to predictions of DBPs' ecological and human health impact based on silico QSAR and MCDM, and advanced and cost-effective water quality sensors will provide real time information on drinking water quality from sources to supply **(EO3, EO4, EO5), delivering ready-made outputs for decision-making and consideration in future legislation**

A central knowledge base with information about DBPs will be fed by authorized users. Alternative drinking water technologies involving MP UV radiation will be targeted in order to reduce the need for chlorination leading to lower levels of DBPs in drinking water,
and innovative and cost-effective treatment approaches involving adsorption and advanced oxidation will be developed to remove contaminants from water **(EO6, EO7, EO8)**

Prevention plans to protect drinking water from the source to the supply will be developed, and project results will be published in an accessible and comprehensive way to the public (**EO9, EO10, EO11), spreading the use of the advanced drinking water treatment and disinfection processes and technologies developed.**

IMPACTS

What are the expected wider scientific, economic and societal effects of the project contributing to the expected impacts outlined in the respective destination in the work programme?

Scientific: Increased understanding of water pollutants and development of new technologies, sensing systems, monitoring infrastructures, drinking water disinfection and water treatment methodologies, satellite work and new algorithms. Scientific innovation will render 20 peer reviewed scientific publications as well as participation in 40 international conferences. 8 PhD and 18 MSc students will be trained in the context of the project, and the project will be included in 4 curricular units at pre- and post-graduation levels. **Economic:** 4 potential innovative technologies/patents and IPR (based on the novel sensing and monitoring system as well as advanced disinfection and water treatment technologies to be developed) and implementation of novel raw water treatments able to remove DBPs and algal blooms precursors which will reduce drinking water disinfection costs by at least 5%. Involvement of other utility companies to spread the integrated monitoring of the water quality using WE network. **Societal:** Enabling the implementation of an integrated monitoring solution that ensures water quality for safe human use across the EU and the integration of the new understanding on prevention and mitigation approaches to DBPs in drinking water at the legislative level. The preventive measures to protect drinking water will target water distributors, stakeholders and consumers which will raise societal awareness and understanding on DBPs. The results will contribute to policy-making through norms, standards and regulation (WP6) in Europe in drinking water and water sources management

13. Quality and efficiency of the implementation

3.1 Work plan and resources

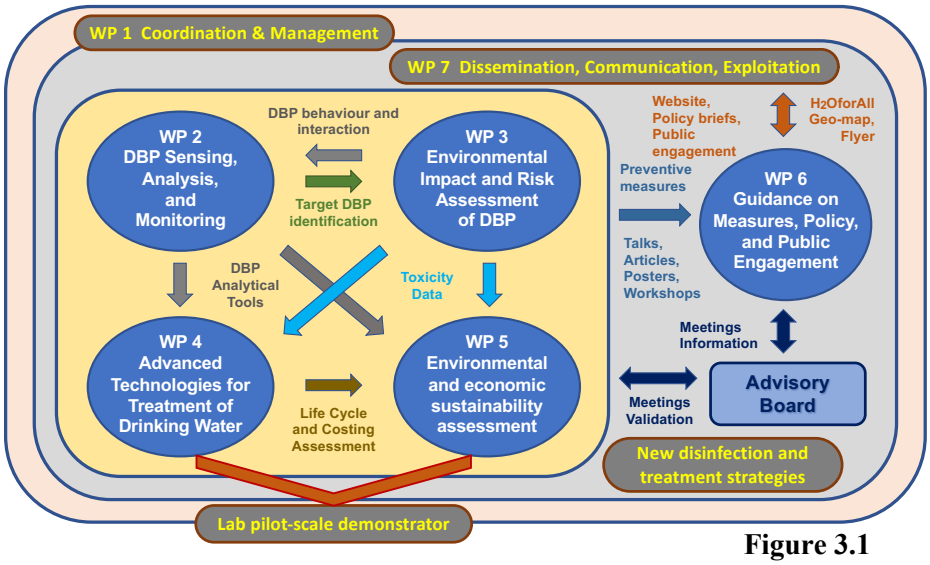

Figure 3.1

H2OforAll aims to innovate targeting to give a strong contribution to halt and prevent pollution, protecting water and drinking water, safeguarding human health and ecosystems, while fostering the European position and role in the global water scene. This involves advanced sensing and monitoring systems (WP2); a deep understanding of contaminants formation pathway, potential impact and interaction with different materials (WP3); innovative technologies for water and drinking water treatment and disinfection (WP4) bearing in mind their environmental and economic sustainability when compared with the standard procedures (WP5); development of preventive measures in an integrated way with several key actors in the water scene while promoting policy guidance and public engagement (WP6) and wide dissemination and communication of the results obtained and exploiting them at different levels to maximize their social, scientific and economic impact (WP7). Figure 3.1 shows the workflow of H2OforAll, giving the interactions between WPs in terms of tools/information/samples/actions dependence. It also provides the bridges for communication beyond the project boundaries. In Figure 3.2, the Gantt chart of the project is presented, indicating the WP leaders as well as the partners responsible for the Tasks.

No.	WP / Tasks designations	WP Task leader & Partners in Task
WP1	**Coordination and Management**	UC
1.1	Consortium management	UC, CYENS, DHV, IVL, Uleeds, SPI
1.2	Administrative management	UC (+All)
1.3	Data Management	UC (+All)
1.4	Project progress and risk management	UC (+All)
WP2	**DBPs Sensing, Analysis and Visualizations**	CYENS
2.1	Optical sensor technologies for water quality monitoring	UULM, UC
2.2	Development of sensing infrastructures for inspection of DBPs	UT
2.3	Cost effective and high-resolution analytical methods for detecting DBPs and relevant materials	Wetsus
2.4	Intelligent placement of the sensing infrastructure along the water distribution network	CYENS
2.5	Design and development of a Central Knowledge Base	IRIS
2.6	Deployment of sensing equipment in the pilot sites	AC
WP3	**Environmental Impact and Risk Assessment of DBPs**	DHV
3.1	Better understanding and assessment of DBPs	CYENS, DHV
3.2	Indices for drinking water safety plan	UC
3.3	Screening and toxicity assessment of DBPs via Multi Criterion Decision Making (MCDM) Tecnhiques	ULeic, UC
3.4	Preventive measures to protect drinking water	DHV, Uhaifa, CYENS
3.5	Hydraulic and quality modelling of water distribution network	AC, UC
WP4	**Advanced and cost-effective technologies for treatment of drinking water**	UC
4.1	New disinfection strategies	Atlantium, Uhaifa
4.2	Source water treatment and DBP removal from disinfected water	UC, Uhaifa, Ugdansk, Uleic, KTH, Adventech
4.3	Optimization of treatment efficinecy by technologies combination in a pilot unit	Adventech, UC, Uleic, UHaifa.
WP5	**Environmental and economic sustainability assessment**	IVL
5.1	Conceptual design and process modelling of a reference case	Adventech, IVL, UC, KTH, Atlantium
5.2	Environmental and economic sustainability assessment of a reference water treatment case	IVL, KTH, Adventech, Atlantium, UC
5.3	Process modeling of DBP prevention and removal technologies implemented to the baseline conceptual design	IVL, KTH, Adventech, Atlantium, UC
5.4	Environmental and economic sustainability assessment of DBP prevention and removal technologies	IVL, KTH, Adventech, Atlantium, UC
WP6	**Guidance on measures, policy and public engagement**	ULeeds
6.1	Citizens awareness and engagement in monitoring and improving water quality	ULeeds , WE, DHV, AC, UC
6.2	Recommendations on Prevention Measures for Safety Planning	DHV, WE, Uleeds, Uhaifa
6.3	EU policy recommendations	WE, Uleeds
WP7	**Dissemination, communication and exploitation**	SPI
7.1	Development and update of the dissemination and communication plan	SPI (+All)
7.2	Development of dissemination and communication materials and digital tools	SPI (+All)
7.3	Knowledge integration, engagement and development of synergies with relevant stakeholders	SPI (+All)
7.4	Development and implementation of the exploitation plan	SPI (+All)
7.5	Links with other relevant actions funded under Horizon 2020 and Horizon Europe calls and targeted topics	WE(+All)

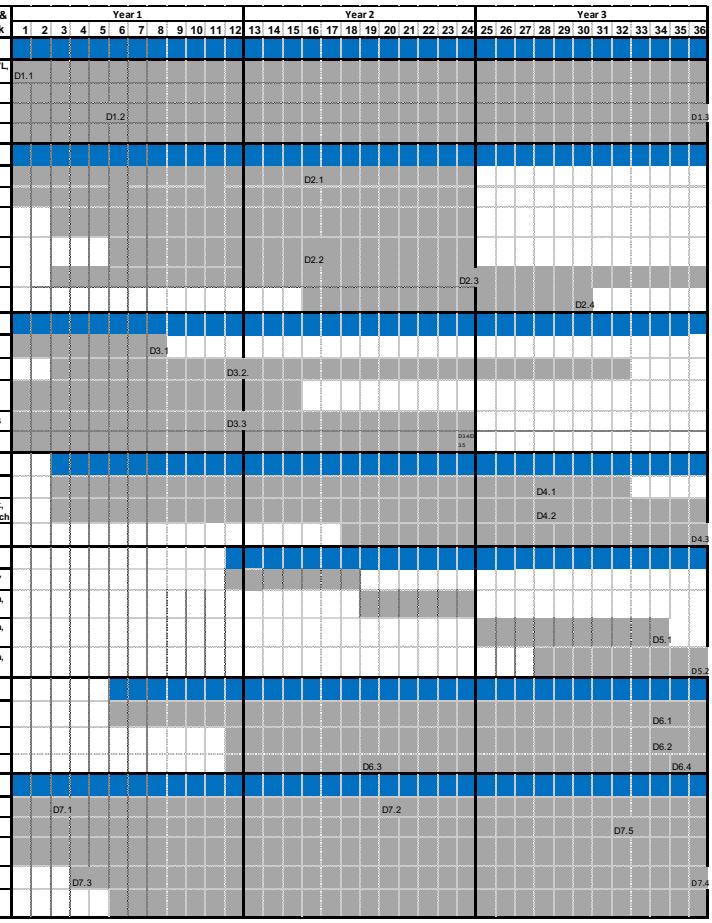

Figure 3.2 - Gantt chart of H2OforAll, including the roles/involvement of partners in WP and Tasks.

Table 3.1. Pilot scale and demonstration summary

Demonstrator	Location	Partner	Scale/Capacity	Start and end TRL	Type and volume of input materials	Type and volume of output materials
Sensing infrastructure	Coimbra	AC	At least 6 sampling points	3 – 5	Water samples	Water quality data
Treatment technologies	Coimbra	Adventech	Lab-pilot unit that will be built to operate in relevant environment	3 – 5	50 – 200 L/h of raw water	50 – 200 L/h of treated water

There is no need for new permit/authorization to competent authorities to carry out these demonstration activities.

Table 3.1g: 'Subcontracting costs' items

13/Atlantium		
	Cost (€)	**Description of tasks and justification**
Subcontracting	8500,00	Research in Task 4.1 includes the measurement of bacterial deposit (biofilm formation) on surfaces, in response to different treatments. To construct the experimental system required for some of these tests, some construction work needs to be done. This work will be done by a contractor with the know-how for pipe-welding (according to Atlantium design).

Table 3.1h: 'Purchase costs' items (travel and subsistence, equipment and other goods, works and services)

1/UC		
	Cost (€)	**Justification**
Travel and subsistence	8852,00	1 project meeting in another partner premises - 3 persons - 2 days = 3000 € 3 conference participation in Europe - 1 person - 3 days = 4000 € Meeting in Brussels with officer - 2 persons - 2 days = 1852 €
Equipment		
Other goods, works and services	57500,00	2 consortium meetings organization = 5200 € (WP1) Reactants for adsorbent and catalysts synthesis / labware / consumables for analytical equipment and for adsorption and AOP tests - 6000 €/ year = 18000 € (WP2; WP4) Analysis outsourcing (electronic microscopies, ssa, etc.) = 2000 €/year = 6000 € (WP4) Open access (3) = 4500 € (WP7) Conference fees = 3 x 600 € = 1800 € (WP7) Printing services = 500 € (WP1) Bioassays – 21500 € (WP3)
Remaining purchase costs	-	
Total	66352,00	
3/UULM		
	Cost (€)	**Justification**
Travel and subsistence	13000,00	2 researchers attending per annual project meeting @ 800 €/person = 1600 € x 3 = 4800 €

		1 researcher attending one technical project meeting per year @ a partner laboratory for collaborative work @ 1200 €/person x 3 = 3600 € 1 researcher attending one international scientific conference per year (e.g. SPIE Photonics West, PITTCON, etc.) @ 1533 €/person x 3 = 4600 €
Equipment	35000,00	1 IR spectrometer module with detector (35000 €, depreciation) – WP2
Other goods, works and services	30000,00	Small optics and electronics for IR sensor (15000 €) – WP2 Chemicals for coatings, MIPs, Calibrations (10000 €) – WP2 Open access publications (5000 €) – WP7
Remaining purchase costs	-	
Total	78000,00	
4/IRIS		
	Cost (€)	**Justification**
Travel and subsistence	11000,00	It includes the attendance of General Assemblies for at least 1-2 people depending on the needs to cover as well as review meetings, attendance of other technical meetings, attendance of workshops as well as installations and facilities visiting)
Equipment		
Other goods, works and services	49200,00	Including desktop component and electronics, cloud hosting, databases, servers and interfaces development tools, security thresholds and licenses) – WP2
Remaining purchase/ costs		
Total	60200,00	
6/UH		
	Cost (€)	**Justification**
Travel and subsistence	6874,60	Project meetings (3) and conferences (2) for 2 people
Equipment		
Other goods, works and services	74375,00	The expenses for measurements of disinfection by products (DBPs) include chromatographic separation columns, reagents, chemicals, eluents, vials. It is estimated that these will add up to a total expense of 12500 € (WP4). Filters and solid phase extraction cartridges is estimated in 5000 € (WP4). Analysis of DBPs will be performed in GC-MS equipment in one of the universities in Israel, based on machine rental per hour. It is estimated that these rentals will lead to a total expense of 13000 € (WP3 and WP4). Other specific water analyses, such as TOC and ion chromatography, will be performed based on machine rental per hour. It is estimated that these rentals will lead to a total expense of 7000 € (WP4). Characterization of the prepared materials for adsorption and oxidation of DBPs by high resolution machines such as XRD, SEM, ICP, zeta potential among others, will be performed in one of the universities in Israel, based on machine rental per hour. It is estimated that these rentals will lead to a total expense of 9500 € (WP4). The budget for consumables and other lab materials such as general chemical reagents, disposables, lab glassware for analytical purposes, centrifuge tubes, tips, maintenance of peristaltic pumps, electrodes etc., is estimated to be 20000 € for the whole project. The expenses for the buildup of lab scale treatment system is estimated in 7375 € (WP4) and will include tanks, connectors, hoses, controllers etc.
Remaining purchase costs		
Total	81249,60	
7/UG		
	Cost (€)	**Justification**

Travel and subsistence	29000,00	Project meetings and conferences: - participation of team members in project meetings (▪ Costs of travel, accommodation, insurance, per diem costs and conference fee for project members participation in international scientific conferences (about 3 000 € per person per conference) – **21 000 €** Cost of travel, accommodation, insurance and per diem costs for project members for participation in project meetings (about 1500 to 2000 € per person per visit) – **8 000 €**
Equipment		
Other goods, works and services	50000,00	Materials and small equipment for MOFs synthesis and characterization (WP4): ▪ Metal precursors for MOFs synthesis (metal salts, such as titanium, copper, zirconium, aluminium and iron precursors; etc) – **4 000 €** ▪ organic ligands for MOFs synthesis (such as: terephthalic acid, trimesic acid, hydroxyisophthalic acid, 2,5-Dihydroxyterephthalic acid, 4,4′-(Hexafluoroisopropylidene)bis(benzoic acid), 4,4′,4″,4‴-(Porphine-5,10,15,20-tetrayl)tetrakis(benzoic acid), 1,2,4,5-Tetrakis(4-carboxyphenyl)benzene, 1,3,5-Tris(4-carboxyphenyl)benzene (prices from 0,08 to 338,33 EUR/gram) – **8 000 €** ▪ solvents for MOFs synthesis and activation (DMF, hexane, ethanol, etc.) – **2000 €** ▪ other reagents for MOFs synthesis for structure controlling (surfactants, ionic liquids, acids, bases, etc) – **2000 €** ▪ small laboratory autoclaves equipped with Teflon lines – **3 000 €** ▪ Laboratory glassware and small laboratory equipment necessary for MOFs synthesis of MOFs and characteristics (magnetic stirrers, hot plates, small laboratory dryer, glass bulbs, Schlenk bulbs, automatic pipettes, laboratory stands, etc) – **12 000 €** ▪ Small laboratory disposable materials for preparation, storage and characterization (gloves, disposable tubes for centrifugation and samples storage, disposable Pasteur pipets, pipets tips, etc) **– 2 000 €** ▪ chemical reagents and small accessories for GC and LC analysis (capillary columns for GC, packed columns for LC, high grade analysis solvents, microsyringes, DBPs standards for analysis, Eppendorf tubes, syringes, glass vials with septa, etc – **5 000 €** ▪ small laboratory equipment for sorption and photocatalytic tests: glass/ quarzt photoreactors, glass set-ups for sorption experiments, connectors, flow meters, laboratory peristaltic pump, Xe arc bulbs, LED diodes, DBPs model compounds for photodegradation and sorption experimenst, etc.) – **12 000 €**
Remaining purchase costs		
Total	79000,00	

8/KTH		
	Cost (€)	**Justification**
Travel and subsistence	13688,00	Conferences and project meetings. It will be 1 project meeting /year. Total 3 project consortium meeting 2 persons from KTH

		joining the project meetings. Travel Cost per participants is 1 200 €/person, subsistence cost is 142 €/person/day, and accommodation cost is 200 €/day/person, Total cost is 11504 €. Also join at least 1 international conference – 1 person. Transportation Cost is 1 200 Euro/person, subsistence cost is 142 Euro/person/day, and accommodation cost is 200 Euro/ person,. Conference fee is estimated 500 Euro/conference. Total cost is 2184 Euro.
Equipment	1961,00	A personal computer will be bought in this project and lifetime for 3 years. – WP4 and WP7
Other goods, works and services	18627,00	Consumables and samples characterization will be used for WP4. Consumables such as N2 gas, calibration gas for all instruments such GC/MS, Micro-GC etc,thermal couples,etc, chemicals used for the catalyst. Estimation cost is 9 313,50 € (WP4) Characterization of samples: a full sample characterization including proximate properties and elemental analysis cost etc. a full sample characterization including proximate properties and elemental analysis cost 200 Euro, at least 14 samples will be analysed Estimation cost is 9 313,50 € (WP4)
Remaining purchase costs		
Total	34276,00	

9/SPI		
	Cost (€)	**Justification**
Travel and subsistence	6000,00	Project meetings 2000 € per meeting
Equipment		
Other goods, works and services	19500,00	Development of final conference in Portugal (10000 €), organization of one project workshop (2000 €); website development (500 €); flyers, brochures and roll-ups (1000 €), 1 project video (6000 €) : WP6 and WP7
Remaining purchase costs		
Total	25500,00	

10/UT		
	Cost (€)	**Justification**
Travel and subsistence	10000,00	Project meetings and conferences; visit to AC during deployment/pilots testing
Equipment	15000,00	WP2: Network interfaces (lora, wifi, 4G) 7000 €, processors (rpi, arduino) 4000 €, gateways 4000 €
Other goods, works and services	2000,00	WP2: casings, batteries, cables, connectors
Remaining purchase costs		
Total	27000,00	

11/ADVENTECH		
	Cost (€)	**Justification**
Travel and subsistence	5000,00	2 people attending 3 presential meetings (1500€/meeting) + Travel to demonstration site in Portugal (500€)
Equipment	30000,00	Stainless steel tanks, piping, accessories (depreciation: 5000 €); Equipment (pumps, automatic valves, pneumatic board, ozone generator) (depreciation:7500€); instrumentation (pH sensor and controllers, level sensors and indicators, oxygen) (depreciation: 8000€); Electric board, cables (depreciation: 5000 €); Reagents (NaOH, H2SO4, coagulant, flocculant) (3500 €); EPIs and other consumables (1000 €) – WP4

Other goods, works and services	15000,00	Programming of control panel (5000 €); Metalworking pilot construction (8000 €); Transport of the unit (2000 €) – WP4
Remaining purchase costs		
Total	50000,00	
12/WE		
	Cost (€)	**Justification**
Travel and subsistence	4800,00	Travel costs for participation to monitoring meetings: 3 project meetings - 2 participants - 800€/travel/participant
Equipment		
Other goods, works and services	14250,00	1 video development (10000 €), printing costs (450 €), organization of a Living Labs event, Involvement in implementation of activities (4000 €). WP6 and WP7
Remaining purchase costs		
Total	19050,00	
13/ATLANTIUM		
	Cost (€)	**Justification**
Travel and subsistence	6000,00	Participation in consortium meeting abroad (one person attending each meeting – 2000 €/person)
Equipment	15000,00	Dedicated UV system is required for tests since the chemicals generated should be tested immediately – 15000 € (depreciation) (task 4.1)
Other goods, works and services	40000,00	1)Lab supply: consumables for bacterial and chlorine tests (12000€) – Task 4.1; 2) Lab supply for bacterial tests (12000€) – Task 4.1; 3) Analytical chemistry tests for radicals generated by UV (16000 €) – Task 4.1
Remaining purchase costs		
Total	61000,00	
16/WETSUS		
	Cost (€)	**Justification**
Travel and subsistence	10000,00	Max. 3 persons, 3 physical meetings, 3 conferences, sensor testing at pilot sites, Collaborative sensor development at partner labs.
Equipment	20000,00	WP2: High-end processors (6000 €); Spectrophotometric sensor (6000 €); Pumps (1800 €); Multichannel transmitter (3000 €); pH sensors (960 €); Conductivity sensors (780 €); ORP sensors (780 €); DO Sensors (680 €)
Other goods, works and services	14000,00	Lab consumables costs: Labware material/Chemicals/Prototyping (5000 €); Electronics (3000 €); Optics (2000 €) Open access journal publications (3 publications – 4000 €) – WP7
Remaining purchase costs		
Total	44000,00	
17/ULEIC		
	Cost (€)	**Justification**
Travel and subsistence	14202,00	7 meetings x 2 people x 918 €/person per meeting = 12852 €; 1 conference = 1350 €
Equipment		
Other goods, works and services	16065,00	Topsis software (2025 €) – WP3, PC for data analysis (1620 €) – WP3, laboratory consumables (5670 €) – WP3 and WP4; Consumables for characterization (4050 €) – WP3 and WP4, open access publications (2700 €) – WP7
Remaining purchase costs		
Total	30267,00	
18/UNIVLEEDS		
	Cost (€)	**Justification**
Travel and subsistence	13750,00	Project meetings and conferences

Equipment	3125,00	Server leasing, smart phones and IoT equipment
Other goods, works and services	15751,28	General consumables
Remaining purchase costs		
Total	32626,28	

3.2 Capacity of participants and consortium as a whole

The consortium reflects a balanced choice of relevant stakeholders in the EU water scene, facilitating the multiactor engagement process. The H_2OforAll consortium involves 18 participants from 10 different countries (Portugal, Spain, Poland, Israel, United Kingdom, Germany, Sweden, Cyprus, Netherlands and Belgium) including 11 Universities/Research Institutes (UC, UULM, ULeicester, CYENS, UHaifa, Ugdansk, KTH, UNIVLEEDS, Utwente, IVL, WETSUS) and 7 Companies from 4 European countries (DHVMED, IRIS Technology Solutions, SPI, Adventech, Water Europe, AtlantiumTLTD, AC). The **research team is composed by 43% female and 57% male investigators**. Several synergies are expected to result from different interactions that will be established between the involved beneficiary and associated partners due to their different and complementary backgrounds, as shown in Figure 3.3.

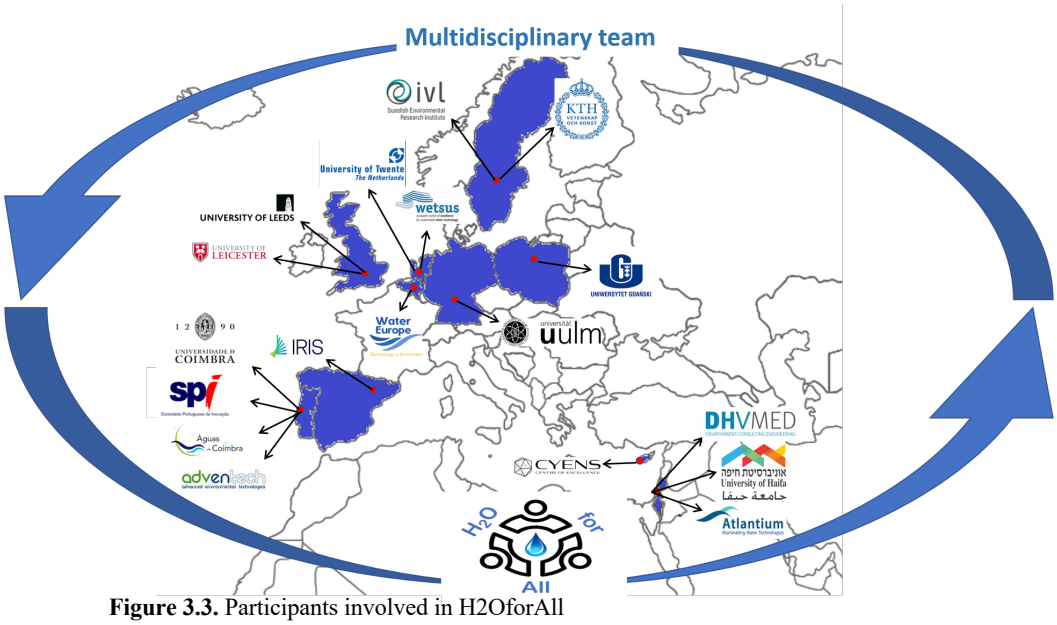

Figure 3.3. Participants involved in H2OforAll

UC is the main proponent of this project and it will be responsible for its coordination and management, in close collaboration with all the participants (WP1). Also, **UC** has a strong background in water treatment technologies development as well as water characterization. **SPI** will work in close collaboration with Researchers from the WE and UNIVLEEDS on the dissemination and communication of the main achievements that will result from each WP and getting people (from academy, industry and the general public) involved in drinking water management issues and pollutants mitigation solutions. **SPI** has previously demonstrated experience in designing and developing materials appropriate and effective for communicating results and outcomes of EC projects to the widest range of relevant stakeholders (participated as a coordinator or partner in over 25 FP7/CIP and 45 H2020 projects). **UNIVLEEDS**, involving social sciences scientists, has large experience in participatory crowd-sensing approaches for increasing citizens' awareness and engaging with policy-making stakeholders (*e.g.* development of collective decision-making platforms such as Smart-Agora, EPOS, *etc.*). **UULM** will bring knowledge in technologies for sensing persistent molecular pollutants in aquatic systems. **IRIS** is an advanced engineering company, certified under ISO 9001:2008 and UNE-EN-ISO 166002, specialized in process monitoring, control and optimization. **UT** has a strong expertise in the development of sensing infrastructures for the monitoring of water quality. **Wetsus** will bring the know-how on the cost-effective and high-resolution methods for detecting the selected molecules and the materials they interact with. **CYENS** develops algorithms and methods to model water distribution networks to detect vulnerabilities to water contamination threats for selecting the most relevant monitoring. Besides this will allow to shade light regarding DBPs formation pathways, fate along the distribution network as well as their interaction with pipes and naturally occurring materials. **DHV** brings the know-how regarding drinking water prevention strategies and regulation. **ULeic** has a strong expertise on toxicity assessment through MCDM and will help prioritizing the most dangerous DBPs as well as to evaluate the novel water treatment/disinfection technologies proposed. The company **Atlantium** has strong expertise in UV disinfection technologies that will complement/substitute the current approaches. **UHaifa** will explore the simultaneous adsorption and oxidation process for pollutants abatement from source water and DBPs removal from drinking water. **KTH** will apply their know-how in magnetic bio-activated carbons from biomass waste to remove contaminants from water while easily recovering the adsorbent. **UGdansk** has large experience in photocatalytic water treatments and photo-materials preparation will be used to degrade contaminants from water and drinking water. Since this project aims the employment of green and sustainable

technologies/solutions/materials within a circular economy concept, specialists from **Adventech** will help to identify synergies between all the materials/processes developed and optimized and select the most suitable ones for scale-up and to build a fully automated transportable lab-pilot scale unit that will be tested under semi-industrial conditions. **IVL** will assess and compare the LCA and LCC of the novel integrated water treatment and disinfection technologies compared with the conventional approaches to reach the most cost-effective and sustainable methodology. **Water Europe** is the recognized voice and promotor of water related RTD and innovation in Europe. With over 250 members across Europe on board, **WE** is recognised by the European Commission as the industry-led European Technology Platform (ETP) for Water. Leveraging from its extensive network and experience in projects, WE will play an important role into the dissemination and policy work packages aiming to maximize the impact of the project activities and results. The utility municipal company **Águas de Coimbra (AC)** manages the water distribution network of Coimbra city and will provide water samples, process water quality data, perform hydraulic and quality modelling through the network, support the field demonstration of the developed sensing infrastructure and promote citizens awareness and engagement on water quality.

The consortium has a high international background (involving countries all around Europe and researchers with high degree of internationalisation) with a multidisciplinary (Chemical, Electro, Civil and Environmental Engineers, Chemistry, Biology, Biochemistry, Sociologists) team with the required skills and infrastructures (as described in Part A of the proposal) for fully develop the workplan proposed. It is worthy to notice that some of the participants' have already been involved in join projects creating bonds of trust and the willingness to work together again and establish new opportunities to advance in science and contribute to an effective transition. An effective share of the critical infrastructures and samples will occur by researchers' interchange among the partners in a cross-fertilizing environment. The participants cover the value chain from the product/technology development, through the relation with citizens, stakeholders and policy-makers all the way to the potential commercialization (bearing in mind the end-users and companies involved in the consortium). All the members of the Consortium were already involved in international and national/regional projects (see selected projects at Part A) involving open science dissemination and have demonstrated expertise and competence in those practices.

An Advisory Board (AB) is constituted for this project, we have already the support of Águas do Centro Litoral that manages the municipal wastewater treatment plants and drinking water production in the Centre Region of Portugal. This company will make part of the AB and will provide water samples for testing the water treatment and disinfection processes that will be developed in H2OforAll. The institute Waterschap Vechtstromen and the Netherlands Institute of Ecology also agreed to make part of our advisory board. Water Europe members will be contacted also to be part of the AB. This board will join the consortium in three of the meetings – M6, M18 and M30, in order to provide feedback about the project achievements and suggest improvements that can increase the impact of the project results. Milestone 1.1 defines month 3 to have the AB constituted with at least 5 institutions. However, the consortium is committed to enlarge the AB with organizations that may bring added-value to the meetings discussions. In these meetings, representatives of other projects in correlated areas can be also invited.

H2OforAll has all the required critical mass, infrastructures, and know-how to fully develop the workplan giving a strong contribution to enhance competitiveness of the EU water sector and fostering the EU's position and role in the global water scene while promoting water protection.

4. Ethics Section

Ethics Issues Table

1. Human Embryonic Stem Cells and Human Embryos			Page
Does this activity involve Human Embryonic Stem Cells (hESCs)?	○ Yes	◉ No	
Does this activity involve the use of human embryos?	○ Yes	◉ No	
2. Humans			**Page**
Does this activity involve human participants?	○ Yes	◉ No	
Does this activity involve interventions (physical also including imaging technology, behavioural treatments, etc.) on the study participants?	○ Yes	◉ No	
Does this activity involve conducting a clinical study as defined by the Clinical Trial Regulation (EU 536/2014)? (using pharmaceuticals, biologicals, radiopharmaceuticals, or advanced therapy medicinal products)	○ Yes	◉ No	
3. Human Cells / Tissues (not covered by section 1)			**Page**
Does this activity involve the use of human cells or tissues?	○ Yes	◉ No	
4. Personal Data			**Page**
Does this activity involve processing of personal data?	○ Yes	◉ No	
Does this activity involve further processing of previously collected personal data (including use of preexisting data sets or sources, merging existing data sets)?	○ Yes	◉ No	
Is it planned to export personal data from the EU to non-EU countries? Specify the type of personal data and countries involved	○ Yes	◉ No	
Is it planned to import personal data from non-EU countries into the EU or from a non-EU country to another non-EU country? Specify the type of personal data and countries involved	○ Yes	◉ No	
Does this activity involve the processing of personal data related to criminal convictions or offences?	○ Yes	◉ No	
5. Animals			**Page**
Does this activity involve animals?	○ Yes	◉ No	
6. Non-EU Countries			**Page**
Will some of the activities be carried out in non-EU countries?	○ Yes	◉ No	
In case non-EU countries are involved, do the activities undertaken in these countries raise potential ethics issues?	○ Yes	◉ No	
It is planned to use local resources (e.g. animal and/or human tissue samples, genetic material, live animals, human remains, materials of historical value, endangered fauna or flora samples, etc.)?	○ Yes	◉ No	
Is it planned to import any material (other than data) from non-EU countries into the EU or from a non-EU country to another non-EU country? For data imports, see section 4.	○ Yes	◉ No	
Is it planned to export any material (other than data) from the EU to non-EU countries? For data exports, see section 4.	○ Yes	◉ No	
Does this activity involve low and/or lower middle income countries, (if yes, detail the benefit-sharing actions planned in the self-assessment)	○ Yes	◉ No	
Could the situation in the country put the individuals taking part in the activity at risk?	○ Yes	◉ No	
7. Environment, Health and Safety			**Page**

Does this activity involve the use of substances or processes that may cause harm to the environment, to animals or plants.(during the implementation of the activity or further to the use of the results, as a possible impact) ?	○ Yes	◉ No	
Does this activity deal with endangered fauna and/or flora / protected areas?	○ Yes	◉ No	
Does this activity involve the use of substances or processes that may cause harm to humans, including those performing the activity.(during the implementation of the activity or further to the use of the results, as a possible impact) ?	○ Yes	◉ No	
8. Artificial Intelligence			**Page**
Does this activity involve the development, deployment and/or use of Artificial Intelligence? (if yes, detail in the self-assessment whether that could raise ethical concerns related to human rights and values and detail how this will be addressed).	○ Yes	◉ No	
9. Other Ethics Issues			**Page**
Are there any other ethics issues that should be taken into consideration?	○ Yes	◉ No	
I confirm that I have taken into account all ethics issues above and that, if any ethics issues apply, I will complete the ethics self-assessment as described in the guidelines How to Complete your Ethics Self-Assessment			☒

Following the advice stated in the Ethics Summary Report, the Consortium acknowledges that some ethical issues might be faced regarding Humans, Personal Data, Non-EU countries and Artificial Intelligence.
The Consortium confirms that compliance with ethical principles and applicable international, EU and national law in the implementation of research activities not originally envisaged (or not described in detail) in the DoA will be ensured. The Consortium also confirms that any ethical concerns raised by those activities will be handled following rigorously the recommendations provided in the European Commission Ethics Self-Assessment Guidelines.